A Semantic Mutation Metric for Metamorphic-Relation Adequacy in Scientific Computing Programs

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Metamorphic testing mitigates the test-oracle problem by checking relations among executions, but classical mutation score is defined over syntactic edits and does not say whether a metamorphic-relation set observes declared domain-semantic effects. This paper introduces Semantic Mutation Score (SMS), an MR-relative adequacy metric over an admitted universe of nonequivalent semantic mutants with explicit certificates and a degeneration path back to classical mutation score. We instantiate five semantic operator families on 12 single-output scientific-computing programs and audit the resulting 60-cell design with an SMS-to-MS proof, an AST-normalized syntactic comparison, aligned-versus-cross relation analysis, and boundary and adjoint studies. The semantic pool has 5.14

CCS Concepts: • **Software and its engineering → Software testing and debugging**; *Software verification and validation.*

Additional Key Words and Phrases: metamorphic testing, mutation testing, semantic mutation testing, metamorphic relation adequacy, metamorphic relation assessment, scientific computing

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1 Introduction

1.1 Software-Engineering Motivation and Scope

Metamorphic Testing (MT) addresses the test-oracle problem in scientific computing software: instead of checking outputs against a known-correct reference, MT checks metamorphic relations (MRs), invariants connecting outputs of related inputs. While *semantic mutation testing* has been named since Clark et al. [13] and tooled for C [18] and for UML state-machine behaviour [22], and while MR-adjacent data-semantic mutation [1, 2, 11, 52] and recent LLM semantic-invariance MT [16] extend the mutation target to test data, relations, and task-level semantics, the adequacy of an MR set against *domain-specific code-level faults*, conservation laws, monotonicity, convergence order, trajectory shape, fidelity ordering, has lacked a backward-compatible metric tied to the classical Mutation Score (MS) lineage. Jia and Harman's [2011] MS is defined over syntactic Abstract Syntax Tree (AST) mutations and does not capture these

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domain semantics. Recent Large Language Model (LLM) generated mutants [30, 46] do produce semantically richer mutants, but do not separate the contribution of LLM source diversity from the contribution of MR design.

Throughout this paper we use the term *four representative classes of single-output scientific computing kernels* (numeric, probabilistic, surrogate, and machine-learning), and avoid the ambiguous software-engineering term *paradigm*. The scope of our claims is strictly bounded to single-output float-to-float kernels: each Program Under Test (PUT) is under 2 KB. We do not claim industrial transfer for multi-output software in this paper.

The four thresholds this study pre-registered for itself, operator instantiability (H1), aligned-versus-cross effect size (H2), cross-class consistency (H3), and kill-attribution purity (H4), are each not met. We report those misses plainly and treat them as the diagnostic result: they are boundary-delimiting findings that locate where operator applicability, MR design, and kill attribution break down, not outcomes to be softened. Against that backdrop, an industrial real-defect arm then tests the same semantic distinction developed here on reproduced defects from widely used scientific-computing libraries: mutation kill-rate, semantic stratum alignment, and real-defect detection are related but different constructs. The arm strengthens the semantic-mutant argument at result level; it is not presented as a reusable benchmark artifact. Its cases were admitted only after defect/MR detectability verification and are reported here at result level. Candidate mining, rejected cases, repository tooling, and benchmark release are outside this manuscript and remain contributions of the archived dataset it draws on.

The intended reader is a software-testing researcher or practitioner who needs to decide whether a set of MRs is adequate for a declared semantic risk, not a domain scientist seeking a new numerical solver or a tool user seeking another LLM-mutant generator. The novelty is therefore the adequacy denominator and certificate structure: SMS asks which admitted semantic-effect fibers are observed by an MR set, while preserving the classical mutation-score form. The practical impact is diagnostic rather than deployment-ready automation: the metric exposes when MR design, operator applicability, or kill attribution prevents a semantic adequacy claim from being trusted.

1.2 Three-layer methodological framework

We organise the contribution as a three-layer framework around domain-semantic mutation operators.

- **Layer 1, Definitional (Section 4.2).** A mutation is *semantic* if it satisfies at least one of three necessary conditions: (a) it crosses a function-call or module-import boundary, (b) it depends on domain knowledge for legality, or (c) it changes the algorithmic class. The five operator families, Conservation Erosion (CE, also written mut_C), Operator Substitution (OS, mut_M), Hyperparameter (HP, mut_G), Trajectory Flip (TF, mut_T), and Structural Injection (SI, mut_F), specialise these conditions across the four PUT classes. We call these five sets *operator families* throughout and reserve *meta-pattern* for the five MR strata (Section 3.3.2); the two are distinct five-element sets placed in one-to-one alignment, so we avoid the label "meta-operator" to keep them apart.
- **Layer 2, Operational (Section 3.3.3, Section 4.10).** An equivalence judgement $E1 \wedge E2$, where E1 is Automated Verification Pipeline (AVP) coherence and E2 is output equivalence on K_eq = 1000 samples, gives the conservative instantiation of the Layer-1 conditions. The trade-off against E1-alone and E2-alone variants is in Appendix A.3.
- **Layer 3, Applied (Section 4.5).** AST-normalised empirical traceability across all 12 PUTs: comparing 292 v4 mutants against 1,250 cosmic-ray syntactic mutants, we show empirically that the v4 pool is not a subset of the syntactic-mutant pool.

Table 1. Evidence-to-claim map for the empirical argument.

Claim supported	Evidence used	Claim not made
SMS is backward-compatible with classical MS	Formal SMS-to-MS degeneration under the syntactic limit	SMS replaces classical mutation score in ordinary syntactic settings
Semantic mutants occupy a different admitted universe	5.14% AST-normalised overlap with default first-order syntactic mutants; HP, SI, and TF are absent from the default first-order pool	Default syntactic mutation tools can never generate higher-order semantic mutants
MR alignment is visible but bounded	Aligned mean SMS 0.213 vs cross mean SMS 0.077 (MP5-held pool); Cliff's δ is positive but below the pre-registered large-effect threshold	The current 12-PUT audit proves a large universal aligned-vs-cross effect
SMS exposes adequacy boundaries	H1–H4 failures, zero-mass cells, root-cause labels, and strong-boundary false-positive / false-negative cases	Failed thresholds are treated as empirical success criteria
Real-defect behaviour is distinct from aggregate kill-rate	The industrial real-defect arm: a pre-registered four-group mutation comparison plus a per-defect detection face on reproduced library defects	The manuscript contributes a reusable real-defect benchmark

The 60-cell empirical audit reported in Section 5 stress-tests this backbone. SMS is positioned as a strict generalisation of classical MS: under the degenerate limit $L = L_{\text{equiv}} \wedge L_{\text{killed}} \wedge L_{\text{mut}}$ formalised in Section 3.3.6, SMS reduces almost everywhere to MS. Layer 1 and Layer 3 correspond to the two leading research questions, RQ1 (constructibility and backward compatibility) and RQ2 (syntactic unreachability); the 60-cell demonstration answers RQ3–RQ5.

The empirical evidence has four roles. The 12-PUT audit is the main stress test of the SMS construction. The AST audit tests whether the semantic-mutant space is reachable by default first-order syntactic mutation. The boundary and adjoint arms map when strong MRs do, and do not, detect semantic effects. The industrial real-defect arm checks whether the construct distinctions remain visible on verified MR-detectable defects. These parts should be read in sequence: the theory defines the metric, the 12-PUT audit stress-tests one empirical instantiation, the boundary and adjoint arms test the strong-MR duality, and the industrial arm checks whether the same construct separation appears outside synthetic mutants and outside author-written kernels.

This table also marks the paper's non-claims. We do not claim that SMS is universally superior to classical MS, that pattern-derived MRs dominate all generic relations, that higher-order mutation has been refuted, that the industrial real-defect arm is a benchmark contribution, or that the reported SMS values define industrial acceptance thresholds.

2 Related Work

Seven lines of prior work bracket the contribution of this paper.

General MT foundations and surveys define the oracle-problem setting and the role of metamorphic relations in paired-test reasoning [12, 40, 45]. Because our empirical subject class is scientific computing, we also take the scientific-software testing survey by Kanewala and Bieman [36] as the nearest domain background.

(a) Classical mutation testing. Jia and Harman [34] and Papadakis et al. [42] survey syntactic mutation. The Coupling Effect Hypothesis (CPH), that tests detecting simple faults also detect complex faults, underpins the use of MS as a fault proxy [6, 21, 35]. We argue in Section 4.5 that CPH holds within syntactic mutation but does not automatically extend across the syntactic-versus-domain-semantic boundary, even when it couples simple syntactic faults to complex syntactic faults. The Section 4.5 empirical evidence (5.14% AST overlap on 12 PUTs, with HP, SI, and

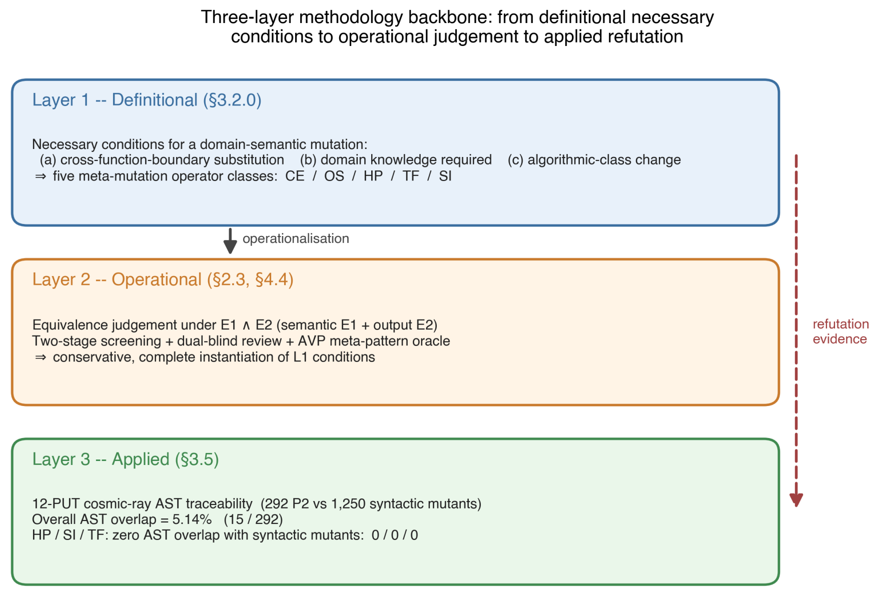

Fig. 1. Three-layer methodological framework: Layer 1 (definitional necessary conditions), Layer 2 (operational $E_1 \wedge E_2$ judgement), Layer 3 (applied AST-normalised traceability across 12 PUTs).

TF at 0/0/0) is the empirical witness for this boundary. The connection between correctness notions and test evidence also precedes modern mutation-score practice [10]; SMS inherits that concern but changes the adequacy object from a program test set to an MR set over admitted semantic mutants.

(b) Higher-order mutation. Jia and Harman [33] and Kintis et al. [37] study compositions of first-order syntactic mutants. We explicitly exclude any equivalence claim about Higher-Order Mutation (HOM) and list HOM equivalence as a residual threat (Appendix F.2).

(c) LLM-generated mutants. Pre-trained-language-model mutation begins with μBERT [19], which masks tokens and uses CodeBERT to synthesise mutants. Tip et al. [46] introduce LLMorpheus, which uses single-LLM JavaScript mutants. Humbatova et al. [30] introduce DeepCrime for deep-learning real-fault mutation. Dakhel et al. [17] extend LLM-driven mutation testing to test generation on Java PUTs, providing an *Information and Software Technology* anchor for the LLM-mutant lineage. To our knowledge no prior LLM-mutant work isolates *LLM source diversity* from *MR design* in the contribution to effect size; the three-stage ablation in Section 4.9 closes this gap. On the equivalent-mutant problem, Delgado-Pérez and Chicano [20] emphasise the importance of the `equiv` term in the MS denominator; we extend that classical bitwise-equivalence definition to a semantic-class equivalence E1 ∧ E2 in Section 3.3.3. Zhang et al. [51] demonstrate MT validation in class-integration test ordering, a non-scientific-computing domain, and the present paper specialises MT to scientific-computing PUTs and couples it to a domain-semantic mutation operator framework. At industrial scale, Meta's ACH [24] deploys mutation-guided LLM test generation to harden test suites against targeted faults on general software; the present paper instead measures MR adequacy on scientific kernels.

(d) Semantic mutation operators across testing communities. Semantic mutation as a research line is older than the LLM-mutant lineage. Clark et al. [13] introduce *semantic mutation testing* as a path distinct from syntactic mutation; Dan and Hierons [18] tool the approach for C; Derezińska and Zaremba [22] apply *semantic mutation operators* to UML state-machine behaviour by perturbing semantic-variation-point interpretations rather than syntactic structure. The distinction matters for our title term: their object is the language interpretation a developer may have misread, whereas ours is a certified violation of a declared domain invariant in code. Within MT specifically, ai Sun et al. [1, 2] propose *data mutation operators* and the μMT methodology to drive MR acquisition through controlled data-semantic perturbation; Zhu et al. [52] formalise *datamorphic testing*, separating datamorphisms (test-case-to-test-case mappings) from metamorphisms (test-case-to-Boolean mappings) over data semantics; Chan and Keung [11] extend generic data mutation operators to unsupervised software-defect-prediction validation. Most recently, Curtò and Zarzà [16] formalise *semantic invariance* in LLM MT under paraphrase, fact-reorder, expand/contract, and context-shift transformations. These works establish that mutation can target data, relation, task, and behavioural semantics. **The contribution of this paper is the first to formalise a domain-specific Semantic Mutation Score (SMS) for MR adequacy in scientific computing kernels, with a proven backward-compatible reduction to classical MS (Theorem 3.1) and a code-level operator framework (Conservation Erosion, Operator Substitution, Hyperparameter, Trajectory Flip, Structural Injection) targeting kernel-specific faults, and to delineate the strong boundary of metamorphic-relation grounding, a strong/weak MR distinction whose ε_{tol}-governed duality (Theorem 3.5, Proposition 3.6) is mapped empirically on real and textbook programs spanning the meta-patterns (Section 5.6), complemented by a cross-library adjoint extension arm (Section 4.7, Section 5.7) that supports the duality forward on third-party linear operators with real fixed defects.**

(e) MR adequacy and coverage. Recent MT-specific adequacy work has begun to ask how MR-based tests should be measured rather than merely generated. The closest literature is best separated by its *counting object*. Xiang et al. [48] construct composite MRs for GSL scientific functions, but evaluate them with the classical MuJava denominator of nonequivalent syntactic mutants. Fu et al. [25] define denominator-bearing MT adequacy over MR obligations and source inputs; Liu et al. [41] integrate MR coverage into MT adequacy; and Ba et al. [7] define Metamorphic Coverage over code elements differentially executed by paired MT runs. These are the closest denominator-bearing MR/MT coverage comparators, but their denominators are MR obligations, source-test obligations, or MR-induced execution coverage rather than semantic mutant classes. Xie et al. [49] quantify MR-set diversity with the MUT model, while Huang et al. [29] and Zhang et al. [50] rank or select MRs using validity, coverage, or mutation-score signals. These latter works are important boundary surrogates for MR-set quality, but they do not define an adequacy fraction over an enumerated universe of semantic fault classes. Blwi et al. [9] provide a useful non-MT analogue by defining test-suite effectiveness through semantic coverage, showing that denominator design can move beyond syntax; however, their adequacy object is a test suite under a specification, not an MR set for scientific kernels. Thus, prior work measures MR exercise, input coverage, differential execution, MR diversity, or syntactic-mutant detection. None makes the adequacy denominator the set of admitted nonequivalent *domain-semantic code mutants* for scientific kernels, and none establishes compatibility with the classical MS denominator. SMS therefore differs not by proposing "another MR coverage metric", but by replacing the denominator against which an MR set is judged.

(f) Property-relative and numerical-specification mutation. Recent property-aware mutation work moves the adequacy question closer to requirements and invariants: Bartocci et al. [8] make mutant relevance and meaningful killing relative to a tested property, and Jeangoudoux et al. [31] generate tests for numerical specifications through interval-constraint reasoning over floating-point accuracy and robustness requirements. These works are closer to

Table 2. Closest denominator-oriented comparators to SMS. The comparison is by adequacy object and denominator semantics, not by whether a paper uses the phrase metamorphic-testing adequacy.

Work	Denominator or counted object	Relation to SMS
[25, 41]	MR obligations, MR coverage, and source-test obligations	Closest MR/MT adequacy comparators, but property/coverage denominators rather than semantic mutant classes
[7]	Differentially executed code elements in paired MT runs	Execution-side coverage comparator, not MR-set semantic adequacy
[49]	MR feature/diversity space	Boundary surrogate for MR-set quality, not an adequacy fraction over faults or semantic classes
[29, 50]	MR validity, coverage, and classical mutation-score signals	Boundary surrogate for cost-effective MR choice, not a denominator-bearing adequacy metric
[9]	Semantic obligations under a specification	Non-MT semantic-coverage analogue; test-suite side rather than MR-set side

SMS than generic AST mutation because they reject property-irrelevant mutants as an adequacy denominator. They still differ from the present paper in three respects: the adequacy object here is an MR set rather than a single formal requirement, the denominator is a five-class domain-semantic mutant space for scientific kernels, and the metric is constructed as a backward-compatible generalisation of MS.

(g) Semantic foundations and the certificate gap. Abstract interpretation supplies the standard machinery for relating concrete semantics to abstract domains and invariant families [14], and for designing program transformations by their effect on an abstract semantics [15]. Approximate program transformation semantics gives a second, quantitative route: local error expressions and partial-metric semantics can certify that a transformation lies within a bounded semantic-effect envelope [26, 47]. Refinement- and UTP-based mutation testing lift mutation to contracts and specifications [3, 4], while usage-aware differential dynamic logic tracks admissible proof-level mutations under soundness conditions [23]. These foundations supply much of the machinery needed for certificate-bearing semantic-effect classes, but they are not MR adequacy metrics for scientific-computing kernels. Conversely, SMS provides the MR adequacy denominator and empirical audit, but its certification layer is empirical and structural rather than machine-checkable proof-carrying.

Closest comparison. The closest adjacent work is the Min-MR-Complete formulation [39], which uses the same semantic-mutation family as the fault model for MR-subset completeness and identifies when class-level reasoning collapses to a mutant-level set-cover problem. The present paper differs in role: it defines the SMS denominator, the E1 $\wedge$ E2 admission judgement, the AST-normalised syntactic-unreachability audit, and the empirical same-source/cross-source ablation. Min-MR-Complete optimises over an admitted mutant–draw coverage universe; this paper constructs and audits that universe. Taken together, the two papers come closest to semantic-effect-class denominators for MR adequacy, but neither yet attaches a per-mutant machine-checkable certificate showing that a syntactic edit belongs to a declared abstract semantic-effect class. Throughout this paper, a certificate is a recorded, budget-stamped piece of test evidence, not a proof object (Limitation 9); the undecidability result is precisely why nothing stronger is available.

The SMS metric is conceptually complementary to the code-verification scope of ASME V&V 20-2009 §3, which targets numerical-solver correctness. SMS targets MR fault-detection adequacy; we make no normative compliance claim (Appendix E.3 records this as a long-term aspiration only).

3 Problem Formulation and Semantic Mutation Model

This section turns the software-engineering problem into auditable research questions before introducing the metric. The order is deliberate: scope and novelty are stated as RQ1–RQ2, while RQ3–RQ5 test whether one empirical instantiation is usable, discriminative, and consistent enough to support the construct. The formal model then gives reviewers a small declared vocabulary, the semantic-certificate record, and the SMS-to-MS degeneration that ties the proposal to the mutation-testing lineage.

3.1 Research Questions

The research questions are ordered from the theoretical foundation (RQ1), through the structural separation that motivates it (RQ2), to the empirical demonstration on the 60-cell audit (RQ3–RQ5). RQ1 is answered by construction and proof (Sections 3.3.7–3.3.9: the undecidability theorem (Theorem 3.3), the fiber characterisation (Proposition 3.4), and the duality theorem (Theorem 3.5), together with the SMS–MS degeneration of Section 3.3.6); RQ2 by the AST-normalised structural audit (Section 4.5); RQ3–RQ5 by the pre-registered statistical pipeline (Section 5). This ordering mirrors the three-layer framework of Section 1.2: Layer 1 (definitional) carries RQ1, Layer 3 (applied) carries RQ2.

- **RQ1 (Theory of semantic mutation: constructibility, undecidability, duality, backward compatibility).** Can semantic mutation be formalised as a fiber decomposition of a single effect map over admitted nonequivalent *domain-semantic code mutants* for scientific-computing kernels, such that (a) stratum membership is undecidable and therefore certificate-based (Theorem 3.3), (b) the induced adequacy metric SMS degenerates almost everywhere to the classical Mutation Score (Proposition 3.4; Theorem 3.1), and (c) strong metamorphic relations detect only the semantically active mutants within a tolerance-bounded regime, while outside it the duality degrades along an ε_{tol}-governed strong boundary into false positives (weak MRs, where a correct discrete program violates the invariant) and false negatives (surviving non-equivalent mutants under coincidental satisfaction) (Theorem 3.5 and Proposition 3.6; the boundary is mapped empirically under RQ4)?
- **RQ2 (Syntactic unreachability).** To what extent is the domain-semantic mutant space induced by the five operators reachable by default-configuration first-order syntactic mutation tools at the AST-normalised level?
- **RQ3 (Instantiability).** Do the five semantic mutation operators instantiate a usable nonequivalent semantic-mutant denominator across the 60 PUT-MP-operator cells, as reflected by inst_rate, equiv_rate, C1_share, and survive_rate?
- **RQ4 (Discrimination, LLM-source decoupling, the strong boundary, and industrial construct separation).** Does SMS discriminate operator-MP aligned slices (j=k) from cross slices (j≠k); does cross-source LLM pooling under an identical prompt move the aligned-versus-cross effect size, or only the kill-set quality; where does the strong boundary of Proposition 3.6 lie empirically, on which real and textbook programs does a strong MR kill the genuine fault, and where does the duality fail as a weak-MR false positive or as a surviving non-equivalent mutant; and does the construct separation among aggregate kill-rate, semantic alignment, and real-defect detection remain visible on an industrial arm of reproduced, MR-detectable defects from widely used scientific-computing libraries?
- **RQ5 (Cross-class consistency).** Does the aligned-versus-cross SMS pattern generalise across four representative classes of single-output scientific-computing kernels? (The relationship between SMS and the coverage-style surrogate Pattern Coverage is reported descriptively at n = 12 as a hypothesis-generating observation, not a confirmatory test.)

3.2 Hypotheses

The empirical questions RQ3–RQ5 carry pre-registered hypotheses; RQ1 is answered by Theorem 3.1 and RQ2 by the Section 4.5 structural audit, neither of which is a statistical hypothesis. The hypothesis definitions, primary-MR convention, and statistical reporting hierarchy are frozen in `docs/experiment_documentation/EXPERIMENT_DESIGN.md` at commit `0f2509527f346f9433c3cf90959bb07d80601a23` and in the Zenodo replication package 10.5281/zenodo.20250664.

- **H1 (RQ3).** At least 4 of the 5 operators produce ≥ 5 non-equivalent mutants on at least 9 of the 12 PUTs.
- **H2 (RQ4; pre-registered as an exploratory large-effect target).** The aligned-SMS to cross-SMS odds ratio is ≥ 3.0, and Cliff's δ is ≥ 0.474. The threshold was fixed before data collection, but the design is underpowered at this boundary, so its verdict is read as a point-estimate result (stipulated-alternative power in Section 5.4), not a confirmatory claim of the paper.
- **H3 (RQ5).** A within-class sign test gives 4/4 across the 4 classes, and CV(ΔSMS) < 0.5.
- **H4 (RQ4).** Mean suspect_share is ≤ 0.20 across the 60 cells.

What would count against the construct (stated post hoc). The pre-registered thresholds stress one instantiation and do not exhaust falsifiability. The construct itself would be disconfirmed by: (i) the degeneration path failing empirically, with SMS and MS disagreeing on the same pools in the syntactic limit; (ii) the aligned-versus-cross direction reversing systematically across designs; (iii) the industrial arm showing the generic baselines' kill sets nesting the pattern-derived kill sets, leaving no construct separation; or (iv) certificates failing audit, with declared strata not matching observed effects. Criteria (ii)–(iv) were observed not to occur; criterion (i) was not directly tested and is open to replication. We state these criteria post hoc, after the verdicts, and label them accordingly.

The 12 PUTs are deliberately *Numerical Recipes*–style compact kernels: they provide a verifiable minimum working example for the three-layer backbone, while industrial-scale transfer is left to future work under the representativeness threat.

3.3 Formal Model and Semantic Mutation Score

3.3.1 Vocabulary and Notation. We use a small vocabulary throughout the method. The terms are declared here before they are used in the SMS construction.

Program Under Test (PUT). A PUT is the scientific-computing kernel being assessed. We write the i-th PUT as S_i.

Metamorphic Relation (MR). An MR is a property that relates the outputs of related inputs. The MR family used for PUT S_i and meta-pattern k is written $\mathrm{MR}_{i,k}$.

Meta-pattern (MP). An MP is a semantic stratum of MR behaviour, such as conservation, monotonicity, convergence, trajectory structure, or fidelity order. The index k denotes the MP being tested.

Semantic operator. A semantic operator is a mutation family mut_j designed to perturb a declared MP. The operator index j identifies the intended semantic effect.

Semantic mutant. A semantic mutant s' is a finite syntactic edit of S_i that is admitted by semantic operator mut_j and is certified against the declared MR family. Formally,

$$s' \in \mathrm{mut}_j(S_i).$$

The word semantic therefore names the certification target, not a second editing mechanism.

Table 3. Operator signatures.

Operator	Failure semantics	Aligned MP
mut_C	Conservation-breaking	MP_1 Conservation
mut_M	Monotonicity-breaking	MP_2 Monotonicity
mut_G	Convergence-breaking	MP_3 Convergence
mut_T	Trajectory-distorting	MP_4 Trajectory
mut_F	Fidelity-order-breaking	MP_5 Partial-order

Semantic-effect fiber. For an MR family $\mathrm{MR}_{i,k}$, the semantic-effect fiber is the subset of admitted semantic mutants whose declared effect is observable by that MR family:

$$\mathcal{F}(\mathrm{MR}_{i,k}) = \{s' \in \mathrm{mut}_j(S_i) \,:\, \mathrm{killed}(s', \mathrm{MR}_{i,k})\}.$$

Different MR families can observe different fibers. SMS therefore measures adequacy relative to a declared stratum; it is not a total order over all possible MR families.

Equivalent, killed, and surviving mutants. A mutant is equivalent for (i, k, j) when the equivalence judgement of Section 3.3.3 admits it into $\mathrm{equiv}_{i,k,j}$. A non-equivalent mutant is killed when at least one MR in $\mathrm{MR}_{i,k}$ passes on S_i and fails on s'. A non-equivalent mutant that is not killed is surviving. Thus the generated mutant set decomposes as:

$$\mathrm{mut}_j(S_i) = \mathrm{equiv}_{i,k,j} \cup \mathrm{killed}_{i,k,j} \cup \mathrm{survive}_{i,k,j} \quad \text{(disjoint)}$$
$$\mathrm{SMS}_{i,k,j} := \frac{|\mathrm{killed}_{i,k,j}|}{|\mathrm{mut}_j(S_i)| - |\mathrm{equiv}_{i,k,j}|} \in [0,\ 1]$$

Semantic Mutation Score (SMS). SMS preserves the classical Mutation Score (MS) ratio while replacing the internal definitions of mutant generation, equivalence, and killing with MR-relative semantic definitions. The classical concepts of mutation operator, mutant, equivalent mutant, killed mutant, surviving mutant, and MS are inherited from Jia and Harman [34] and Ammann and Offutt [5].

The complete notation table is in Appendix A.1; index sets and tolerance parameters ($\varepsilon_{\mathrm{eq}}, \varepsilon^k_{\mathrm{AVP}}, K_{\mathrm{eq}} = 1000, N = 20$) follow that table.

3.3.2 *Semantic-Operator Signatures.* Operator family and alignment:

$$\mathrm{MUT} = \{\mathrm{mut}_C,\ \mathrm{mut}_M,\ \mathrm{mut}_G,\ \mathrm{mut}_T,\ \mathrm{mut}_F\} \quad \text{(open, extensible)}$$
$$\mathrm{mut}_j \,:\, \mathrm{Programs} \to 2^{\mathrm{Programs}}$$
$$\mathrm{align}(j) = j \quad \text{(design choice, not theorem)}$$

Per-PUT specialisation tables (mut_C/M/G/T/F across 12 PUTs and 4 classes) are deferred to **Appendix B.2**.

3.3.3 *Equivalence Judgement.* The equivalence judgement is the executable instantiation of the Layer-1 necessary conditions (Section 4.2). For each candidate mutant `s'`:

- **E1 (AVP-coherent):** ∀ mr ∈ $\mathrm{MR}_{i,k}$: $\mathrm{AVP}(S_i, mr) = \mathrm{AVP}(s', mr)$.
- **E2 (Output-equivalent):** ∀ x in $K_{\mathrm{eq}} = 1000$ samples $\sim D_S$: $\|S_i(x) - s'(x)\| \le \varepsilon_{\mathrm{eq}}$.

$E1 \wedge E2$ is the conservative complete instantiation: false-equiv (a truly non-equivalent mutant admitted as equivalent) requires *both* AVP coherence and numerical agreement on K_eq samples to hold despite the non-equivalence (both mis-pass simultaneously), which is rare; false-non-equiv requires only one of E1, E2 to fail for a truly equivalent mutant, biasing SMS slightly high. The trade-off table against E1-alone and E2-alone is in Appendix A.3. Under the Section 3.3.6 degenerate limit `L`, $E1 \wedge E2$ reduces almost everywhere to classical bitwise equivalence (Lemma G.1, supplementary Appendix G).

Semantic certificate record. For each admitted mutant, the empirical certificate is a structured record rather than a proof-carrying object. The record contains the PUT identifier, operator identifier, intended meta-pattern stratum, edit class, targeted semantic invariant, E1 AVP-coherence result, E2 sampling budget and tolerance, killed/survived vector over the relevant MRs, LRCA labels for killed outcomes, generation source and reviewer status, and a trace path to the reproduced artifact. A mutant enters the SMS denominator only after the record passes the Layer-1 semantic-condition check and the $E1 \wedge E2$ equivalence judgement. LRCA labels remain diagnostic annotations on the certificate; they do not filter the denominator or redefine SMS.

The killed determination preserves the classical OR-aggregation:

$$\begin{aligned}\mathrm{killed}(s', \mathrm{MR}_{i,k}) \iff \exists\, mr \in \mathrm{MR}_{i,k} &: \\ &\mathrm{AVP}(S_i, mr) = \mathrm{pass} \wedge \mathrm{AVP}(s', mr) = \mathrm{fail}.\end{aligned}$$

3.3.4 Root-Cause Diagnostic Layer. Likely Root-Cause Analysis (LRCA) annotates every killed mutant with one of five diagnostic labels: C1 likely semantic failure, C2 numerical-tolerance perturbation, C3 out-of-distribution (OOD) trip, C4 statistical-assumption violation, and C5 mutator artefact. The diagnostic procedure is a three-layer decision tree. Layer 1 checks tolerance robustness over N = 20 repeats with a fail-ratio cutoff of 0.80. Layer 2 triages OOD on the surrogate and machine-learning classes. Layer 3 checks statistical-assumption baselines on the probabilistic and machine-learning classes using Wilcoxon and Dynamic Time Warping. A final pass rechecks for mutator artefacts. Multiple labels are resolved by priority C5 > C4 > C3 > C2 > C1.

The output quantities are `C1_share` (share of C1 among killed mutants) and `suspect_share := 1 − C1_share`. **LRCA does not modify the SMS formula**: the killed set is never filtered by suspect status. LRCA is a diagnostic annotation that indicates whether a given kill is more consistent with a semantic-failure detection, tolerance sensitivity, OOD behaviour, statistical-assumption failure, or mutator artefact. Appendix A.2 gives the full decision tree, the 9-grid threshold calibration, and the engineering rationale for the priority ordering.

3.3.5 Backward-compatibility declaration. The seven core concepts (PUT, mutation operator, mutant, equivalent mutant, killed mutant, surviving mutant, mutation score) and the SMS formula `|killed| / (|mut| − |equiv|)` are aligned item-for-item with Jia and Harman [34]. We extend only the internal definitions: `mut` shifts from syntactic to domain-semantic operators, `equiv` shifts from bitwise behavioural equivalence to the semantic-class equivalence $E1 \wedge E2$, and `killed` shifts from an equality oracle to a meta-pattern AVP. We introduce no new formula terms and no new state classifications.

Under the degenerate limit L defined in Section 3.3.6, SMS reduces almost everywhere to classical syntactic MS, equivalent mutants degenerate to classical behavioural equivalence, killed mutants degenerate to bitwise difference detection, and the LRCA layer trivialises, C2 through C5 cannot fire when $L1 \wedge L2 \wedge L3$ hold. Any SMS-based empirical conclusion is therefore structurally consistent with the existing mutation-testing literature in the classical syntactic-mutation regime, so there is no metric-level semantic fragmentation. The skeleton of eleven concepts, the three-state decomposition, the SMS formula, and the AVP interface are fixed; the contents of MUT, MP, C, and cls remain open to future extension.

3.3.6 Degeneration to Classical Mutation Score. The Section 3.3.5 backward-compatibility claim is formalised as a degeneration theorem. The full notation cross-reference is **Appendix G.1**; the joint conditions are in **Appendix G.2**, the lemma proofs in **Appendix G.3**, and the full theorem proof in **Appendix G.4**. We state only the main theorem and corollary here; the three construction lemmas are stated and proved in the supplementary material as Lemma G.1–G.3.

Definition (Degenerate limit). $L = L_{\text{equiv}} \wedge L_{\text{killed}} \wedge L_{\text{mut}}$, where each joint condition is a pair of paired axes acting on one layer of the SMS formula:

- $L_{\text{equiv}} = L1 \wedge L2$: $\varepsilon_{\text{eq}} \to 0 \wedge K_{\text{eq}} \to \infty$ (controls equiv layer; Lemma G.1).
- $L_{\text{killed}} = L3 \wedge L4$: $\varepsilon_{\text{AVP}} \to 0 \wedge$ MP set = $\{\text{MP}_{\text{eq}}\}$ (controls killed layer; Lemma G.2).
- $L_{\text{mut}} = L5 \wedge L6$: mut_j switches to rule-based syntactic operators (Mothra-style, or modern MutMut/PIT-style) $\wedge$ PUT class $\subseteq$ imperative deterministic programs (controls mut layer; Lemma G.3).

Theorem 3.1 (SMS → MS degeneration). *In the degenerate limit L, **almost everywhere** with respect to the input distribution D_S,*

$$\text{SMS}_{i,k,j} \xrightarrow{L} \text{MS}_{i,j} = \frac{|\text{killed}_{i,j}^{\text{classic}}|}{|\text{mut}_j^{\text{syntax}}(S_i)| - |\text{equiv}_{i,j}^{\text{classic}}|}$$

where the right-hand side is the classical Mutation Score of Jia and Harman [34], $\text{killed}^{\text{classic}}$ *is the difference-detection set,* $\text{equiv}^{\text{classic}}$ *is the behavioural-equivalence set, and* $\text{mut}^{\text{syntax}}$ *is the syntactic-mutant set.*

Proof sketch. By Lemma G.1, equiv_{i,k,j} → equiv_{i,j}^classic almost everywhere with respect to D_S; the measure-zero qualifier accommodates floating-point pathological points and Not-a-Number propagation. By Lemma G.2, killed_{i,k,j} → killed_{i,j}^classic, and the MP index k degenerates because L4 collapses MR_{i,k} to {MP_eq}. By Lemma G.3, mut_j(S_i) → mut_j^syntax(S_i). Substituting into the SMS formula gives MS_{i,j}. Full lemma proofs are in supplementary Appendix G.3 and the full theorem proof in supplementary Appendix G.4. ☐

Under the stated substitutions the equality is definitional once the three layers coincide; the almost-everywhere qualifier concerns only the bounded E2 classifier on floating-point pathological inputs. We therefore read Theorem 3.1 as a backward-compatibility characterisation of SMS, not as an independent mathematical contribution.

Corollary 3.2 (LRCA trivialisation). *Under L, the likely-root-cause inventory C = {C1, ..., C5} degenerates to {C1}, so suspect_share → 0 and SMS becomes a single-layer metric consistent with the engineering attribution structure of Jia and Harman [34].*

Empirical consistency. Theorem 3.1 and Corollary 3.2 jointly show that the SMS-based empirical conclusions (Section 5 Cliff's delta, Friedman chi^2, Spearman rho) are structurally consistent with existing mutation-testing literature in the classical syntactic-mutation regime, and **do not** constitute metric-level semantic fragmentation.

3.3.7 Semantic-Mutation Theory Setting. The preceding SMS-construction subsections fix the SMS metric and its degeneration to MS. We now give the theory the metric instantiates: semantic mutation is *intensionally semantic, extensionally syntactic*. Every mutation is mechanically a syntactic act, because semantics changes only through syntax. What makes a mutant *semantic* is where its defining condition lives (at the semantic layer, as a specified invariant violation) and how its membership is established (by certificate, because the condition is undecidable).

Fix a grammar G and the set $\mathrm{Prog}(G)$ of well-formed programs with denotational semantics $[\![\cdot]\!] : \mathrm{Prog}(G) \to (\mathcal{X} \rightharpoonup \mathcal{Y})$. Let $\alpha : \mathcal{Y} \to \mathcal{Y}_\alpha$ be the committed semantic abstraction (the observable that the AVP of Section 3.3.3 reads), and write $P \equiv_\alpha P'$ when $\alpha \circ [\![P]\!] = \alpha \circ [\![P']\!]$ pointwise on the admitted input domain $\mathcal{X}_{\mathrm{adm}}$. This $\equiv_\alpha$ is exactly the equivalent-mutant relation equiv of Section 3.3.1: the E2 component of the E1∧E2 judgement (Section 3.3.3) is its bounded decision procedure on K_{eq} samples.

The new ingredient is a finite *semantic invariant family* $I = \{\psi_1, \dots, \psi_5\}$, where each ψ is a predicate over the observable behaviour $\alpha \circ [\![P]\!]$. The five invariants are exactly the properties the meta-patterns of Section 3.3.2 target: ψ_1 conservation, ψ_2 monotonicity, ψ_3 convergence order, ψ_4 trajectory determinism, ψ_5 partial-order consistency, so ψ_k is the invariant whose violation MP_k detects and whose breakage mut_j (at $j = k$) targets. We write $[\![P]\!] \vDash \psi$ when P's observable behaviour satisfies ψ.

The family is open (the extensibility of I noted in Section 3.3.9): a sixth invariant ψ_6, *adjoint consistency* $\langle Ax, y\rangle = \langle x, A^{\mathrm{H}} y\rangle$ for linear-operator kernels, instantiates the same construction. Its MR is strong (its violation set is closed under $\equiv_\alpha$), so by Theorem 3.5 it detects only the semantically active mutants of the adjoint code path. Because adjoint consistency is meaningful only for linear operators , not for the chaotic, stochastic, and nonlinear machine-learning kernels that make up most of the 12 PUTs, ψ_6 is *not* folded into the pre-registered 60-cell design (doing so would leave the invariant undefined on ten of the twelve subjects); it is instead demonstrated as a separate confirmatory *adjoint extension arm* on dedicated third-party linear-operator subjects (Section 4.7, Section 5.7), leaving the pre-registered hypotheses H1–H4 untouched.

3.3.8 Semantic-Mutant Conditions. The following certified definition is the formal counterpart of the informal Layer-1 necessary conditions of Section 4.2.

Definition (semantic mutant). Given (P, ψ) with $[\![P]\!] \vDash \psi$, a program P' is a *semantic mutant of P at stratum* ψ iff: (S1) *realizability*, $P' = e(P)$ for a finite AST edit e; (S2) *baseline*, $[\![P]\!] \vDash \psi$; (S3) *violation with witness*, $[\![P']\!] \nvDash \psi$, witnessed by some $x \in \mathcal{X}_{\mathrm{adm}}$ observable through α; (S4) *latency*, P' compiles, terminates on $\mathcal{X}_{\mathrm{adm}}$, and is type-correct, so the defect is observable only at the semantic layer, never by a crash oracle; (S5) *stratum purity* (required where stratum labels feed downstream), $[\![P']\!] \vDash \psi'$ for all $\psi' \in I \setminus \{\psi\}$.

A semantic mutation class is therefore defined by which invariant must break and how (S2–S5) and realized by whichever syntactic edits achieve it (S1). The operators $\mathrm{mut}_C, \dots, \mathrm{mut}_F$ of Section 3.3.2 are the constructive templates for strata $\psi_1, \dots, \psi_5$.

Definition (latency window). For a violation-magnitude parameter ε of an edit template, the *latency window* is the interval $(\varepsilon_{\mathrm{tol}}, \varepsilon_{\mathrm{crash}})$: above the tolerance at which the ψ-checker witnesses the violation (S3), below the threshold at which the container crashes (S4). An empty window certifies non-realizability of the template at that site.

3.3.9 Effect map, fibers, and three theorems. **Definition (effect map and fibers).** For fixed P, the *effect map* σ sends an applicable edit e to the semantic-effect class of $P_e = e(P)$, valued in

$$\{\ \equiv_\alpha,\ \text{ill-formed},\ \psi_1\text{-viol}, \ldots, \psi_5\text{-viol},\ \text{active-off-taxonomy}\ \},$$

where active-off-taxonomy means $P_e \not\equiv_\alpha P$ yet no $\psi \in I$ flips from satisfied to violated. The *fiber* of a class is its σ-preimage. P_e is *semantically active* iff $P_e \not\equiv_\alpha P$; the strict syntactic-only mutants are exactly the inactive fiber $\sigma^{-1}(\equiv_\alpha)$. σ is single-valued only on the *S5-pure sub-domain* $\{e : |\{\psi \in I : [\![P_e]\!] \not\models \psi\}| \leq 1\}$; on the corpus this sub-domain covers 263/292 (90.1%) of admitted mutants (the S5 audit of Section 5.9). On the residual 9.9%, σ is a relation rather than a function, and those edits are carried explicitly as multi-stratum and excluded from the pure-fiber reading.

Theorem 3.3 (undecidability of semantic-mutant membership). *For Turing-complete* $\mathrm{Prog}(G)$ *and non-trivial* ψ*, deciding whether an edit e is a semantic mutant at stratum* ψ *(conditions S2–S4) is undecidable.*

Proof sketch. S3 requires deciding $[\![P_e]\!] \not\models \psi$, a non-trivial semantic property, undecidable by Rice's theorem; the termination clause of S4 is independently undecidable by reduction from the halting problem. Semantic-mutant status therefore cannot be read off the edit; it can only be certified on bounded fragments with recorded budgets, which is the formal reason the admission judgement (Section 3.3.3, Section 4.10) is gate-shaped rather than a decision procedure. □

The theorem is a routine consequence of Rice's theorem, in the same spirit as the classical undecidability of mutant equivalence; we state it to fix the formal reason admission is gate-shaped, not as a novel result.

Proposition 3.4 (fiber characterization of classical practice). *(i) Every semantic mutant arises from some syntactic edit; the stratum-*ψ *class equals* $\sigma^{-1}(\psi\text{-viol})$ *intersected with S4. (ii) The classical equivalent-mutant problem is exactly the degenerate fiber* $\sigma^{-1}(\equiv_\alpha)$ *of unconstrained syntactic sampling. (iii) Classical mutation testing and semantic mutation are the same effect map under two sampling policies: sample the edit domain blindly, versus sample a specified non-degenerate fiber.*

Proof. (i) holds because semantics changes only through syntax; (ii) and (iii) follow by unfolding the definitions. □

The SMS-to-MS degeneration of Theorem 3.1 (Section 3.3.6) is the metric-level shadow of (ii): when the admitted fiber collapses to the syntactic default, the SMS denominator collapses to the MS denominator.

Theorem 3.5 (duality with strong MRs). *Let r be a metamorphic relation whose violation set is closed under* $\equiv_\alpha$ *(a* strong *MR). Then r detects no semantically inactive mutant, and conversely any mutant detected by a strong MR is semantically active. Hence over a strong MR universe the kill matrix is supported exactly on the semantically active fibers, and inactive (syntactic-only) mutants are structural noise for strong MT evaluation rather than hard cases.*

Proof. If $P_e \equiv_\alpha P$ and r's violation predicate factors through α, then r flags P_e iff it flags P; MR validity says r does not flag P, so it does not flag P_e. The converse is the contrapositive. □

The duality identifies α as the single hinge connecting MR-side quality (strength) and mutant-side quality (activity). Classification is relative to $(I, \alpha, \text{budget})$: extending I can only move a mutant from active-off-taxonomy to ψ-stratified, never the reverse, and bounded $\equiv_\alpha$ verdicts are search conclusions, not proofs, so every certificate carries its $(I, \alpha, \text{budget})$ stamp. The full reduction for Theorem 3.3 follows Rice's theorem together with the standard halting reduction for the termination clause; the proof of Theorem 3.5 is the $\equiv_\alpha$-closure argument above.

Definition (strong MR, weak MR, strong boundary). Theorem 3.5 reads off strength algebraically, assuming the invariant ψ behind r is satisfied by the correct program so that closure under $\equiv_\alpha$ makes a violation diagnostic. For a discrete program P approximating a continuous operator this assumption is *contingent*: the correct P reproduces the operator's algebraic invariant only up to a discretisation residual, so $\equiv_\alpha$ between P and the ideal solution holds only within the tolerance ε_{tol} of the latency-window definition. Fixing ε_{tol}, call r *strong* on P when the correct P satisfies r within ε_{tol} (its residual stays below ε_{tol} under refinement), so that any violation beyond ε_{tol} certifies a semantically active mutant and Theorem 3.5 holds operationally; call r *weak* on P when the correct P's residual exceeds ε_{tol}, so that the correct program is itself flagged because the discretisation, not a fault, breaks the invariant. The *strong boundary* is the $(\varepsilon_{\text{tol}}, \text{resolution})$ locus separating the two regimes; tightening ε_{tol} or coarsening the discretisation carries a strong MR across it into the weak regime. A weak MR is exactly a non-strong (non-grounded) MR in the effective, discrete sense: its violation set is not closed under the program's ε_{tol}-level $\equiv_\alpha$.

Proposition 3.6 (the two boundary failures and their ε_{tol} coupling). *Relative to $(\varepsilon_{\text{tol}}, P)$ the duality of Theorem 3.5 has two failure modes. (i)* False positive (weak MR). *When r is weak the correct P violates r and a non-fault is reported; the rotational-symmetry invariant of a neutron-transport operator under a method-of-characteristics discretisation, satisfied only at special angles, is the canonical instance. (ii)* False negative (surviving non-equivalent mutant). *A semantically active mutant $P_e \not\equiv_\alpha P$ can leave every invariant in I satisfied within ε_{tol}, a* coincidental satisfaction of the MR, *and so survive the strong MR family; this surviving mutant is non-equivalent (genuinely $\not\equiv_\alpha P$) and is therefore distinct from the classical equivalent mutant of Proposition 3.4(ii), and when its signature lies below ε_{tol} it is a* tolerance fault. *The two modes are coupled through ε_{tol}: tightening it shrinks the false-negative set (more surviving mutants are killed) but grows the false-positive set (more correct programs are flagged), and loosening it does the reverse. The boundary is therefore an ε_{tol}-governed* sensitivity–specificity trade-off, *not a defect to be engineered away.*

Proof. (i) and (ii) instantiate the Definition: weakness places the correct program's residual above ε_{tol}, and coincidental satisfaction places the mutant's signature below it; monotonicity of both sets in ε_{tol} holds because both membership predicates are threshold comparisons against the same ε_{tol}. □

Operationally, mode (i) does not inflate SMS: the killed predicate requires every counted MR to pass on the unmutated program, so a weak MR is rejected upstream at MR validation and contributes no kills. Proposition 3.6(i) therefore manifests as MR-validation failure on the correct program, and only mode (ii) enters the SMS denominator-numerator arithmetic directly.

Theorem 3.5 thus holds within the strong regime, and the ε_{tol}-governed strong boundary, where a strong MR degrades into a weak MR (false positives) or is evaded by a surviving non-equivalent mutant (false negatives), is mapped empirically on real and textbook programs in Section 5.6.

4 Study Design

The study design follows three validation paradigms rather than relying on a single small experiment. First, analytical validation establishes that SMS is a conservative extension of classical MS. Second, the controlled 12-PUT, 60-cell audit tests constructibility, syntactic separation, aligned-versus-cross discrimination, and root-cause attribution under a reproducible protocol. Third, boundary, adjoint, and real-defect evidence check whether the construct behaves as expected outside the main synthetic-mutant grid. The design is intentionally bounded: compact kernels make semantic certification inspectable, while industrial transfer is treated as a future validity question rather than assumed.

4.1 Program Selection

Table 4. PUT selection.

Class	PUT	Name	Mathematical structure	LOC
A Numeric	A1	Lorenz ODE integration	Nonlinear ODE	~150
	A2	LU decomposition	Linear algebra	~80
	A3	FDM 1D heat conduction	Parabolic PDE	~200
B Probabilistic	B1	Beta-Binomial conjugate	Analytic posterior	~60
	B2	MCMC Metropolis-Hastings	Markov chain	~250
	B3	Monte Carlo integration	Importance sampling	~100
C Surrogate	C1	Gaussian Process Regression	Kernel methods	~300
	C2	Polynomial Chaos Expansion	Orthogonal basis	~250
	C3	Neural-net surrogate	MLP substitution	~400
D ML	D1	Multi-Layer Perceptron	Backprop.	~350
	D2	Support Vector Machine	Convex optimisation	~200
	D3	Logistic Regression	Maximum likelihood	~120

The 12 PUTs are taken from the shared 12-PUT infrastructure [39] but independently justified along four dimensions: library-stack coverage (numpy 2.4.4, scipy 1.17.1, scikit-learn 1.8.0); mathematical-structure coverage (8 of 12 chapters of *Numerical Recipes*); overlap with existing mutation-testing benchmarks (DeepCrime, Defects4J, and the mutmut and cosmic-ray demos); and signature-simplification trade-offs. The full coverage argument is in Appendix B.1. The `program(x: float) → float` signature is a substantive constraint (Section 6) that bounds the upper limit of mutant semantic complexity; transfer to industrial multi-output PUTs is left to outside the evaluated scope of this paper.

4.2 Necessary conditions for semantic mutation

Definition (Semantic mutation criteria). A mutant `s' = mut_j(S_i)` is a *semantic* mutation if and only if it satisfies at least one of the following three conditions.

(1) **Cross-function-boundary replacement.** The AST node operated on crosses at least one function-call or module-import boundary. For example, `np.linalg.det(M) → np.sum(np.diag(M))`.

(2) **Carries domain knowledge.** The legality of the mutation depends on mathematical, physical, or statistical knowledge of the program's domain, not on purely syntactic type preservation. For example, changing the Gaussian Process Regression `noise_level` from 1e-4 to 1e-1.

(3) **Changes algorithmic class.** The mutation alters the algorithmic class implemented. For example, replacing RK4 with Euler changes the integration order.

A mutation that satisfies none of (a) – (c) is purely *syntactic*: AST-local, domain-agnostic, and class-preserving. The five operator families (CE, OS, HP, TF, SI) are specialisations of (a) – (c):

Table 5. Necessary conditions for semantic mutation.

Operator class	(a)	(b)	(c)	Primary condition
CE Constant perturbation	×	△	×	partial (b); weakest
OS API replacement	✓	✓	△	(a)+(b)
HP Hyperparameter	×	✓	△	(b)+partial(c)
TF Numerical transform	△	✓	✓	(b)+(c)
SI / CF Structural injection	△	✓	✓	(b)+(c)

Only **CE partially satisfies** the necessary conditions (it is a semantic / syntactic boundary class); OS, HP, TF, SI strongly satisfy at least one of (a), (b), (c).

4.3 Metamorphic-Relation Coverage Matrix

Table 6. Metamorphic-relation coverage matrix: per-PUT coverage density across the five meta-patterns (12 PUTs × 5 meta-patterns). •• substantial (32 cells), • moderate (19 cells), ∘ vacant (9 cells; no MR is exercised).

PUT	Conservation	Monotonicity	Convergence	Trajectory	Partial order
A1 Lorenz	••	•	••	••	∘
A2 LU	••	∘	•	•	••
A3 FDM	••	•	••	••	∘
B1 BetaBin	••	•	∘	∘	•
B2 MCMC	•	••	••	••	•
B3 MC	••	∘	••	•	∘
C1 GPR	•	••	••	•	••
C2 PCE	••	•	••	•	••
C3 NN-Surr	•	••	••	••	••
D1 MLP	••	••	•	•	••
D2 SVM	•	••	•	∘	••
D3 LR	••	••	•	∘	••

Legend: •• substantial (32 cells); • moderate (19 cells); ∘ vacant (9 cells; no MR exercised).

Each PUT averages 24.3 mutants in the cross-source pool (range 10-30), for 292 unique v4 mutants across the 12 PUTs; each mutant is evaluated in its relevant cells over the 60-cell design with N=20 AVP repetitions per cell, and K_eq

= 1000 input samples for E2 evaluation. Detailed pool counts and per-class operator specialisations are in **Appendix B.2**.

The cell density notation distinguishes ••substantial (32 cells; aligned slice + strong off-diagonal cells where MR coverage is dense and mutant detection is expected), •moderate (19 cells; cross slice with non-trivial expected detection rate), and ∘ vacant (9 cells; historically empty cells where no MR is exercised); the counts are read mechanically from the coverage grids recorded in the MR implementation files (32 substantial, 19 moderate, 9 vacant). The 12 aligned (j = k) cells are defined by each PUT's class-primary meta-pattern; the 48 cross (j ≠ k) cells comprise the remaining substantial, moderate, and vacant cells. The •moderate / ••substantive distinction does not change SMS computation; it tracks expected MR coverage density per the infrastructure documentation and informs the Section 5.9 R_sem / R_kill decoupling discussion.

Per-class operator specialisations (illustrative). Each operator family (mut_C / M / G / T / F = CE / OS / HP / TF / SI in their dual form) requires PUT-class-specific specialisation:

- **mut_C Conservation-breaking:** A1 Lorenz adds $\varepsilon_{\text{drift}}$ to RHS (slow Hamiltonian drift); A2 LU decomposition omits the (k+1)-th row multiplier; B1 Beta-Bin posterior omits normalisation; C1 GPR covariance omits positive-definite diagonal term; D1 MLP backprop omits one gradient term.
- **mut_M Monotonicity-breaking:** A3 FDM Δt occasionally negative; B2 MCMC acceptance min(1, r) → min(0.95, r); C2 PCE high-order coefficient sort inserts inversion; D2 SVM decision-function sign flips near boundary.
- **mut_G Convergence-breaking:** A1 Lorenz RK4 → 1.5-order hybrid; A3 FDM 2nd-order difference → 1st-order; B3 MC doubling sample size does not 1/N-reduce variance; C3 NN-Surr training-epoch truncation.
- **mut_T Trajectory-distorting:** A1 Lorenz state-vector y / z swap; B2 MCMC inserts independent-sampling segment; C3 NN-Surr training-target slow phase shift; D1 MLP hidden-layer periodic-mask activation.
- **mut_F Fidelity-order-breaking:** A2 LU partial pivoting degrades to no pivoting; C1 GPR length-scale switches to coarse prior; C2 PCE high-order term randomly retains low-order; D3 LR regularisation occasionally large.

The per-class HP / OS / TF substitution rules give similarly differentiated specialisations across PUT classes (e.g., HP on class a = tolerance / max_iter, on class c = GPR `noise_level` / `length_scale`, on class d = MLP `hidden_dim` / dropout; OS on class a = numerical-linalg API swap `det` ↔ `sum(diag)`, on class b = sampling-API swap, on class c = surrogate-class swap GPR ↔ RBF ↔ NN). The full PUT-class × operator specialisation grid is in Appendix B.2.

4.4 Primary Meta-Pattern Convention

The diagonal cells `j = k` form the H2-aligned slice, the off-diagonal cells form the cross slice, and the vacant cells ∘ are not formally adjudicated.

The c-class primary MP is held at the pre-registered choice (the partial-order meta-pattern) throughout this paper. All H1, H2, H3, and H4 verdicts are rendered under this configuration on v3 (same-source) and v4 (cross-source under an identical prompt). An earlier exploratory data-driven primary-MP shift was considered but withdrawn from the analysis after a permutation null over fully exchangeable c-class (PUT, MP) cell SMS values showed the resulting δ inflation to be statistically indistinguishable from random reselection of the c-class primary MP; the corresponding pre-registration of a primary-MP-selection rule on a fresh dataset is deferred to a follow-up study. The null pools all 15 c-class cell SMS values, shuffles them over the three PUTs and five MR labels, recomputes the max-over-five primary

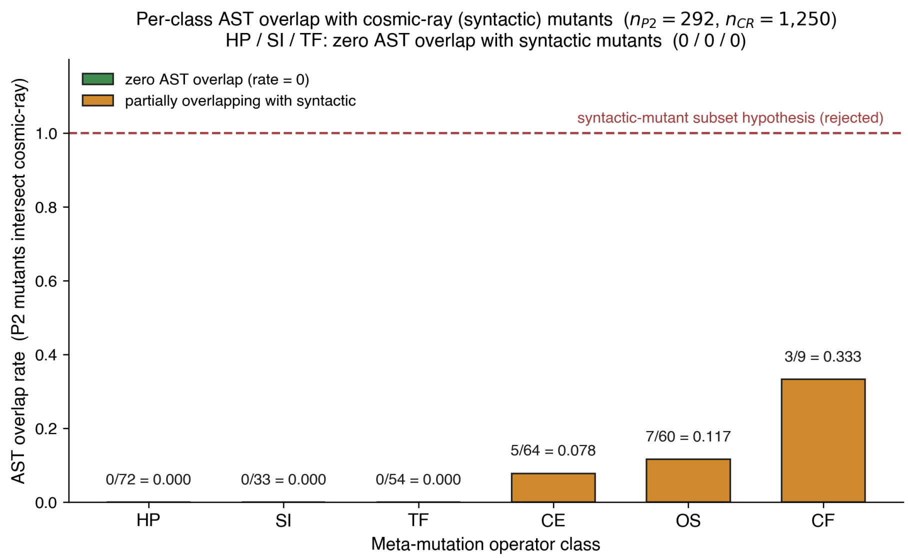

Fig. 2. Per-class AST overlap rate between 292 semantic mutants and 1,250 cosmic-ray syntactic mutants across 12 PUTs (overall 5.14%). HP, SI, TF (54.5% of the semantic pool) show zero overlap (0/72, 0/33, 0/54): SI and TF are structurally cross-function, while HP zero overlap reflects the default first-order value menu; CE 7.81%, OS 11.67%, CF 33.33% (CF is a class-specialisation, not in main 5).

choice per PUT, and averages the selected values. With 10,000 permutations (seed 42), the observed c-class aligned mean is 0.314, the null mean is 0.349, and the one-sided permutation probability of a null value at least as large as observed is 0.989 (`data/results/c_class_permutation_v4.json`).

4.5 Abstract-Syntax-Tree Overlap against Syntactic Mutants

Experimental design. For each PUT, semantic cross-source mutants (`data/mutants/${PUT}_pool_v4/`, 292 total) are compared at the AST-normalised level (`ast.dump(annotate_fields=False, include_attributes=False)`) with cosmic-ray default-operator mutants (1,250 total; `scripts/p2_vs_syntactic_ast_diff_batch.py`). Source: `data/results/cosmic_ray_12put_ast_diff.json`.

Aggregate results.

Table 7. Semantic vs syntactic mutants: 12-PUT empirical.

Metric	Value
Total semantic mutants (12 PUTs)	292
Total cosmic-ray syntactic mutants (12 PUTs)	1,250
AST-normalised overlap	15
Overall overlap rate	**5.14%**

Per-operator-class breakdown (12-PUT aggregate).

Table 8. Semantic vs syntactic mutants: 12-PUT empirical.

Class	n_p2	n_overlap	Rate	Interpretation
HP	72	0	**0.000**	Value-menu artifact (not structural)
SI	33	0	**0.000**	Structurally cross-function
TF	54	0	**0.000**	Structurally cross-function
CE	64	5	0.078	Boundary class (Section 4.2 partial (b))
OS	60	7	0.117	Partial incidental hits
CF	9	3	0.333	b2 only, n=9

Interpretation: refuting the "post-classification copy" challenge.

- HP, SI, and TF (159 of 292 = 54.5% of v4 mutants) show **zero overlap** with default first-order configurations of cosmic-ray's AST-local operators such as BinOp, Compare, and NumberReplacer. For SI and TF the gap is structural: they are cross-function edits that no first-order AST-local operator can express. For HP the zero overlap is a value-menu artifact of the default first-order operator set, an AST-local single-literal edit whose specific replacement value the default menu does not enumerate, rather than a structural impossibility. Whether HOM-based compositions [32] (not refuted) can close the gap is a residual higher-order mutation threat, addressed conceptually in Section 4.6.
- The CE class shows 7.81% incidental overlap, mostly LLM-generated half-step perturbations on a2 and b3 such as `_RHO=28.0 → 27.5`. The remaining 92.19% of CE mutants are AST-disjoint, because the LLMs prefer domain-aware perturbations over integer ±1 changes.
- An earlier draft labelled the OS row "× tool-inexpressible". The 12-PUT data refines this categorical claim to "△ 88.33% disjoint, 11.67% incidental hits". The systematic-versus-incidental argument in Section 4.6 (with details in Appendix B.1.5) clarifies why those incidental hits are stochastic byproducts rather than systematic semantic mutation.
- 94.86% of the v4 mutants cannot be reproduced by cosmic-ray defaults. The two pools occupy systematically distinct mutant spaces.

A multi-tool cross-comparison was not run because mutpy is incompatible with Python 3.10+, and mutmut's operator set overlaps strongly with cosmic-ray's. The DeepSeek, Claude, and GPT contributions to the 15 overlap files are uneven (DeepSeek 11, Claude 4, GPT 0). Appendix B.3 covers the cosmic-ray a1 single-PUT pre-12-PUT pilot, and Appendix F discusses the LLM-source distributional-shift threat.

4.6 Higher-Order Mutation Boundary

Scope of the claim. The preventive-defence claim below is conditional on a first-order syntactic baseline; HOM-based syntactic compositions are an open question and are not refuted by the Section 4.5 evidence.

The HP, SI, and TF zero-overlap result, combined with the Section 4.2 necessary-conditions argument, amounts to a *preventive-defence* argument: semantic mutation operators address a class of fault hypotheses that lie beyond the reach of default first-order syntactic configurations (HOM not refuted; see the scope note above). Three points sharpen this framing.

(a) Systematic versus incidental. Satisfying conditions (a), (b), or (c) is sufficient for a single semantic mutation, but only when satisfaction reflects design intent, not stochastic byproduct, does it constitute a *systematic* semantic mutation method. A syntactic tool that occasionally hits (a) or (c) with its 13 default operators does so at a non-zero probability, but the hits are not repeatable and they carry neither of the two engineering goods we want. First, designing a semantic mutator like OS `det` → `sum(diag)` requires knowing that these expressions are equivalent on diagonal matrices but not on general matrices, a deepening of source-code understanding the syntactic tool does not exhibit. Second, domain-semantic faults, wrong physical constants, unit conversions, boundary conditions, hyperparameter semantics, numerical-method order, are not AST-local; the syntactic-tool design goals (operator typos, off-by-one errors, negation flips) do not target them. Systematic semantic mutation therefore requires (a), (b), or (c) to be design intent. Appendix B.1.5 records this argument in full.

(b) HOM caveat. HOM [33, 37] could in principle compose syntactic mutants, for example, an Arithmetic Operator Replacement combined with a Statement Deletion, to partially simulate the effects of OS or HP on some PUTs. The Section 5 empirics did not run a HOM comparison; we list HOM equivalence testing as residual threat higher-order mutation equivalence (Appendix F.2) and confine our "tool unreachability" claim to *first-order* syntactic tools (mutmut and cosmic-ray default configurations).

Conceptual analysis of HOM reachability. Higher-order mutation (HOM; 32, 33, 37) composes multiple first-order operators on the same source location. A natural challenge to this paper's preventive-defence framing is whether HOM compositions of cosmic-ray's 13 default operators or mutmut's 14 default operators can reach the Section 4.2 necessary conditions (a) cross-function-boundary, (b) domain-knowledge-dependent, or (c) algorithmic-class-change. We address this conceptually without empirical HOM testing.

For (a): HOM compositions remain over first-order operators that are AST-local within their target node; chaining AOR ∘ Statement-Deletion within a single function does not introduce a cross-module substitution like `det(M)` → `sum(np.diag(M))`, which presupposes that `np.diag` is in scope and semantically equivalent on diagonal matrices but not on general matrices. HOM can reach a kindred destination only if both operators happen to coincide with a meaningful cross-boundary swap, which is exponentially unlikely under the default operator menus.

For (b): HOM has no awareness of which constants are load-bearing in the PUT class. A two-operator chain such as Number-Replacer (1e-4 → 1e-3) ∘ Boolean-Swap on a Gaussian-process kernel does not "know" that 1e-4 is the regularization knob; it modifies the literal numerically without crossing the domain-semantics boundary. The Hyperparameter operator class (HP, mut_G) is by construction parameter-aware and class-specific, a property HOM compositions do not exhibit by default.

For (c): Algorithmic-class change (e.g. polynomial → spline basis; Lorenz RK4 → 1.5-order hybrid) requires structural code rewrites at the level of function bodies, library imports, or iteration schemes. HOM compositions over default first-order operators are bounded above by AST-local edits and so cannot reach algorithmic-class change without being effectively equivalent to a manual rewrite.

We therefore assert that the Section 4.5 0/0/0 unreachability for HP / SI / TF, while strictly empirical only for first-order configurations, is unlikely to be refuted by default-operator HOM compositions on combinatorial grounds. A direct empirical HOM falsification (generating mutmut second-order mutants and re-running the AST overlap analysis) remains a residual higher-order mutation threat.

A multi-syntactic-tool cross-comparison (mutmut and mutpy) was not run: mutpy is incompatible with Python 3.10+, and mutmut's operator set strongly overlaps cosmic-ray's. The full operator-class cross-reference between cosmic-ray (13 default operators) and mutmut (14 default operators), including which Section 4.2 necessary conditions each operator family reaches, is in Appendix B.6. The aggregate of the two default sets (~21 distinct first-order AST-local operator classes) collectively fails all three Section 4.2 necessary conditions.

(c) Source distributional shift. The 15 cosmic-ray AST overlaps are not evenly distributed across the three LLMs (DeepSeek 11/15, Claude 4/15, GPT 0/15; see `cosmic_ray_12put_ast_diff.json`). DeepSeek tends to generate syntactically simpler mutations. This LLM-source bias is discussed as a distributional-shift threat (Appendix F.2) but does **not** affect the systematic-vs-incidental argument, which is based on *categorical* AST-locality, not hit frequency: the HP / SI / TF zero-overlap result is 0/0/0 across all three sources.

4.7 Adjoint Extension Subjects

The adjoint invariant ψ_6 (Section 3.3.7) is exercised on a separate, confirmatory set of three third-party linear-operator libraries: scipy `LinearOperator` (linear algebra), pylops real-FFT (signal processing), and jax `linear_transpose` (automatic differentiation), disjoint from the 12 PUTs and the pre-registered 60-cell. Each library carries a real fixed defect in its adjoint code path as a ground-truth anchor (scipy #8900, pylops #292, jax #6223). For each, the MR families are adj.self (self-adjointness) and adj.dual (scaled/dual-operator transpose), and the mutant pool mixes adjoint-breaking mutants (the real defect among them) with semantically equivalent mutants a strong MR must *not* kill; the real defects are reproduced in extracted-diff form because the pre-fix releases are uninstallable on the host (no `arm64` wheel). The arm is confirmatory and additive: it does not alter the 60-cell denominator, the primary-MP convention, or the pre-registered hypotheses. Full subject specifications and fault mechanisms are in supplementary Appendix H; results in Section 5.7.

4.8 Empirical Study Procedure

With the subjects, operators, and the adjoint extension arm fixed (Sections 4.1–4.7), we now specify the procedure that produces each cell's measurements.

Pipeline order: Section 4.1 PUTs → Section 4.9 Mutant generation → LRCA L0 prescreen → Section 4.11 Equivalence ($E_1 \wedge E_2$) → AVP → killed/survive → LRCA three-layer diagnosis → SMS + C_1_share report.

Each (i, k, j) cell is reported with `inst_count`, `equiv_count`, `killed_count`, `survive_count`, SMS_{i,k,j}, C1_share_{i,k,j}, suspect_share_{i,k,j}, and the three rates `inst_rate / equiv_rate / survive_rate` corresponding to RQ3.

4.9 Cross-Source Generation Protocol

To isolate the contributions of LLM same-source bias and MR-MP alignment design to Cliff's delta, we ran a three-stage ablation:

Table 9. Cross-source protocol.

Version	Mutant pool	c-class primary MP	Use
v3	Same-source (Claude Opus 4.6)	partial-order (pre-registered)	H2 baseline (pre-registered primary)

Table 9. Cross-source protocol.

Version	Mutant pool	c-class primary MP	Use
v4	**Cross-source** (Claude Opus 4.6 + GPT-5.4 + DeepSeek chat)	partial-order (held at pre-registered)	Isolate LLM-source diversity

The v4 protocol (`scripts/cross_source_campaign.py`) runs each (PUT, operator) pair on Claude / GPT / DeepSeek with K=3 trials under an identical prompt template (temperature 0.7, V1-V4 mechanical-validation gate). Cross-source pool capacity: 37 operators × 3 sources × 3 trials = 333 attempts, 89.5% V1-V4 pass, 298 confirmed mutants distributed nearly equally (Claude 101, GPT 98, DeepSeek 99); final-stage rejection of duplicate and QA-failing mutants leaves the 292 that enter the v4 pool used for SMS and the AST audit. Here "37 operators" denotes the executable operator-specialisation inventory: 36 primary PUT-class operator pairs plus one CF structural-injection specialisation. The conceptual operator family count remains five (CE, OS, HP, TF, SI); CF is reported only as a specialised SI/structural-injection row in the AST-overlap audit.

Declared confound: protocol asymmetry. v3 used the original Phase-1 dual-blind reviewer protocol (Claude generation + GPT-5.4 review + DeepSeek arbitration); v4 passes V1-V4 mechanical gates only and does **not** invoke a reviewer LLM. A fraction of the v3 → v4 source-axis contrast may therefore reflect a slight quality shift rather than LLM-source diversity per se. The follow-up study will rerun dual-blind on the v4 grid, but at $n \geq 30$ PUTs: a same-n rerun removes the protocol-asymmetry confound yet, at $n = 12$ PUTs, the $\Delta\delta$ CI remains ≈ 0.44 wide, so detecting a $\Delta\delta = 0.20$ source-diversity effect at 80% power requires ≈ 30 PUTs, and the follow-up is therefore designed at that size. Full per-LLM token / latency / cost figures, V1-V4 specifications, and the full differential-prompt protocol are in **Appendix C.1**.

4.10 Mutant-pool prescreen and equivalence detection

Pool prescreen (LRCA L0). Each candidate mutant passes three gates:

(1) Static lint and type checking.
(2) A unit self-test on simple inputs to confirm that the PUT loads, runs, and returns a finite output.
(3) A double-blind review sign-off under the Phase-1 protocol.

In Phase-1, the generator LLM is Claude Opus and the reviewer LLM is GPT-5.4; the two roles are isolated, and the reviewer sees only the PUT and the mutant code and outputs a (syntactic, executable, fault-injected) triple. Inconsistent reviewer outputs go to a manual arbitration queue (no more than 10% of cases); double-confirmed mutants enter the pool. We then randomly sample 20% of the pool for manual review by scientific-computing researchers, and downgrade the entire batch to manual review if manual–reviewer inconsistency exceeds 10%. Each cell produces 30–50 candidates and retains 10–15 mutants. The v4 cross-source pool uses a different L0 execution path: it applies V1–V4 mechanical validation, duplicate removal, and final QA rejection before SMS evaluation, but it does **not** invoke the reviewer LLM, for cost and speed reasons. Consequently v4 C5 (artefact) labels are assigned only after a killed mutant reaches LRCA artefact recheck, where a human reviewer inspects the mutant code, generation trace, and prompt history for over-injection or other mutator artefacts; they are not inherited from a blind LLM reviewer vote. A fraction of the v3 →

v4 source-axis shift may therefore reflect a quality decline rather than a source-diversity contribution. We declare this protocol asymmetry in Section 5.8; full justification is in Appendices C.1–C.3.

Equivalence detection (E1 $\wedge$ E2). For each mutant $s' \in \mathrm{mut}_j(S_i)$:

(1) **E2 step.** Sample $K_{\mathrm{eq}} = 1000$ inputs $x \sim D_S$; compute $\|S_i(x) - s'(x)\|$; if $\leq \varepsilon_{\mathrm{eq}}$ for all samples, mark E2-pass.
(2) **E1 step.** For every $mr \in \mathrm{MR}_{i,k}$, compare $\mathrm{AVP}(S_i, mr)$ with $\mathrm{AVP}(s', mr)$; if all consistent, mark E1-pass.
(3) **Conjunction.** E1 $\wedge$ E2 $\rightarrow$ enter $\mathrm{equiv}_{i,k,j}$, exclude from SMS denominator. Otherwise pass to AVP-killed determination.

E2 is run before E1 because E2 is computationally cheaper (K_eq scalar evaluations) and quickly discards numerically distinct mutants; the remaining candidates are screened by AVP for MR-relation equivalence. The joint condition is conservative: false-non-equivalent (E1 $\vee$ E2 fail) is much easier than false-equivalent (both must mis-pass simultaneously), biasing SMS slightly *low*: truly equivalent mutants misclassified as non-equivalent remain in the admitted denominator as unkillable survivors, understating adequacy, an explicit conservative engineering choice (Appendix A.3).

Killed determination. With OR-aggregation across mr in MR_{i,k}:

$$\mathrm{killed}(s', \mathrm{MR}_{i,k}) \iff \exists\, mr \in \mathrm{MR}_{i,k} \ : \\ \mathrm{AVP}(S_i, mr) = \mathrm{pass} \wedge \mathrm{AVP}(s', mr) = \mathrm{fail}.$$

Each (i, k, j) cell is sampled N = 20 times (statistical replicate) to compute the per-cell SMS, with mutant-level fail ratios feeding the LRCA L1 decision. The AVP version is pinned by the frozen dependency file and the embedded source in the replication package; the reproducibility package carries the complete AVP source so that the package remains self-consistent under any upstream evolution.

4.11 Three-Layer Root-Cause Diagnosis

For each killed mutant the three-layer decision tree applies in order:

- **L1 tolerance robustness** (all classes). N=20 replicates with a fail-ratio cutoff at 0.80; a kill that survives less than 80% of the 20 replicates is labelled C2 (numerical-tolerance perturbation) and exits.
- **L2 OOD triage** (C / D classes only). Sample from D_S^{valid}; if the mutant fails *only* in the OOD region, label C3 (out-of-distribution) and exit.
- **L3 statistical-assumption baseline** (B / D + Wilcoxon / DTW only). Pre-check IID / stationarity on the PUT's own repeated samples; if the AVP statistical assumption is itself violated, label C4 and exit.
- **Artefact recheck**. An external reviewer re-examines mutant code + prompt history; if LLM / artefact evidence (e.g., mutator over-injection) is found, label C5; otherwise label C1.

Multi-label priority C5 > C4 > C3 > C2 > C1 takes the earliest confirmed non-semantic cause (decision-tree control flow). Threshold calibration over a 9-grid (`ood_band` $\in \{0.02, 0.05, 0.10\} \times$ `tolerance_multiplier` $\in \{3.0, 10.0, 30.0\}$, repeats fixed at 20) lifts H4 from 10/60 to 12/60 cells; the best combination (`ood_band = 0.02, tolerance_multiplier = 3.0`) is reported as primary, with default-threshold results retained as control (`lrca_60cell_v3.json`). The calibration ceiling (12/60 = 20%) remains far below the 80% pre-registered threshold; H4 is unattainable on this dataset as an intrinsic property of LLM-mutant pools, not a calibration issue. Detailed grid table, L0-L3 sub-protocols, and decision-tree pseudocode are in **Appendix A.2 + C.4**.

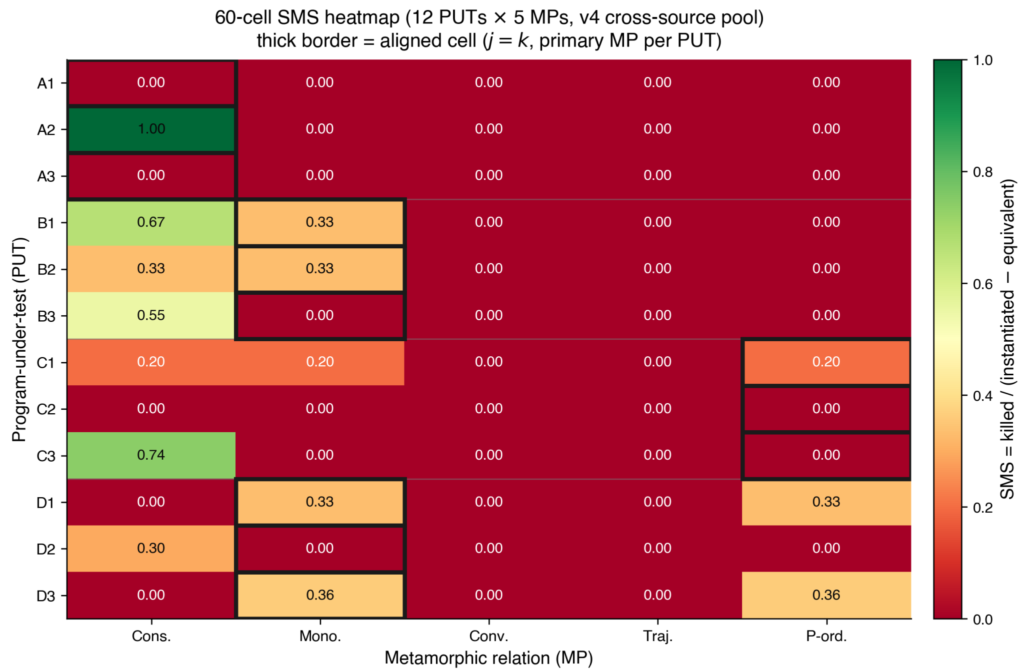

Fig. 3. 60-cell heatmap of mean SMS over 12 PUTs × 5 MPs (v4 cross-source primary). Diagonal cells (operator-MP aligned) show concentrated mass; off-diagonal mass reflects bleed into adjacent MPs. Empty rows = PUT-level zero-mass cohort (PUTs A1, A3, C2; see Section 5.9).

5 Results and Discussion

The empirical thresholds H1–H4 are stress tests of this particular semantic-mutant instantiation, not prerequisites for the definition of SMS. The positive validation target is narrower and more structural: SMS must preserve the mutation-score denominator, separate domain-semantic mutants from default first-order syntactic mutants, make MR adequacy failures diagnosable, and distinguish mutant kill-rate, semantic-stratum alignment, and real-defect detection. Thus the failed thresholds below are reported as boundary findings about operator applicability, MR design, and kill attribution; they do not invalidate the metric construction.

5.1 Statistical pipeline

The primary statistical reporting follows a pre-registered hierarchy:

Table 10. Statistical pipeline.

RQ	Primary statistics	Reporting format
RQ3	inst_rate, equiv_rate, C1_share, survive_rate	60-cell heatmap + 4-class marginals
RQ4	aligned-SMS vs cross-SMS; sparse ∘ vs dense ••equiv_rate	Cliff's delta + odds ratio + 95% bootstrap CI

RQ5	ΔSMS_c (c ∈ {A,B,C,D}); CV(ΔSMS)	directional consistency count over the 4 classes (no p-value attached) + descriptive forest plot
RQ5 (desc.)	Spearman ρ + Kendall τ (SMS vs PC)	scatter + dual-metric ranking comparison

This pipeline answers the empirical questions RQ3–RQ5; RQ1 is answered by Theorem 3.1 (Section 3.3.6) and RQ2 by the AST-normalised structural audit (Section 4.5), outside the statistical pipeline.

Where multiple per-class tests are reported (the per-class Friedman tests of Section 5.11), they carry a Bonferroni correction across the four classes; we do not report a separate cell-level family of p-values, so no family-wise cell-level correction is claimed. Bootstrap 95% CIs use 1,000 iterations as default (B = 10,000 for the headline H2 delta CI). Mixed-effects modelling (`sms ~ C(class) * C(operator) + (1 | put)`) failed with a Singular matrix at N = 60: the class × operator interaction design matrix is rank-deficient and the PUT random-intercept variance collapses to the boundary at this sample size. The Friedman test is the non-parametric formal alternative. The four hypotheses are pre-registered: H1 (operator implementability) requires ≥ 4 of 5 operators producing ≥ 5 non-equivalent mutants on $\geq 9/12$ PUTs; H2 (aligned-vs-cross) requires odds ratio ≥ 3.0 *and* Cliff's delta ≥ 0.474 (Romano 2006 large-effect threshold); H3 (cross-class consistency) requires sign-test 4/4 across 4 classes plus CV(ΔSMS) < 0.5; H4 (LRCA mass) requires mean suspect_share ≤ 0.20. SMS variants (SMS, SMS_unfiltered) and construction lemmas are in **Appendix D.1**.

The results are reported in RQ order. RQ3 is answered in Section 5.2. RQ4 spans the aligned-versus-cross comparison (Section 5.3), its power analysis (Section 5.4), the root-cause diagnostics (Section 5.5), the strong-boundary and adjoint studies (Sections 5.6 and 5.7), the cross-source analysis (Sections 5.8 and 5.9), and the industrial construct-separation arm (Section 5.10). RQ5 follows in Sections 5.11 and 5.12, with the class-level stability discussion in Section 5.13.

Because the 60 cells share 12 PUTs and are not independent, each reported statistic carries an explicit inference permission (Table 11); no number in this paper should be read above its row.

Table 11. Inference permissions for the reported statistics.

Statistic	Pre-registration status	Sampling unit (n)	Licensed reading
H1 operator–PUT counts	pre-registered threshold	PUT × operator	deterministic count; verdict factual on this pool
H2 Cliff's δ, odds ratio, bootstrap CI	pre-registered exploratory large-effect target	cell (12 aligned vs 48 cross; non-independent within PUT)	point-estimate verdict only; no population claim (pass probability ≈ 0.5 at the boundary is rule-intrinsic; Section 5.4)
H3 sign test, CV(ΔSMS)	pre-registered threshold	class (n = 4)	verdict factual on these four classes
H4 mean suspect_share	pre-registered threshold	cell (n = 60)	descriptive mean; no independence assumed

Friedman χ^2 (MP effect; per-class Bonferroni)	exploratory inferential	PUT blocks (n = 12)	non-parametric alternative after mixed-effects singularity; exploratory
Spearman ρ / Kendall τ (SMS vs PC)	none (descriptive)	PUT (n = 12)	hypothesis-generating only
Industrial arm: four-group Wilcoxon, Holm-corrected	pre-registered in the dataset protocol	case (n = 34, paired)	inferential within the 34-case pool; no corpus-level claim
Industrial arm: 34/34 real-defect face	none (descriptive; selection-conditioned)	case	face-level count; not a coverage rate

5.2 RQ3: Semantic-Mutation Score Distribution

Table 12. RQ3: Semantic-mutation score distribution.

Metric	Value
Number of cells	60
Mean SMS	0.104
Median SMS	0.000
Std SMS	0.213
Cells with SMS = 0	45 / 60
Mean mutants / PUT (v4)	24.3 (range 10-30)

The **zero-mass dominance** (45/60 = 75%) is concentrated in the cross-MP slice: ~88% (42/48) of cross cells are zero, ~25% (3/12) of aligned cells are zero. Cliff's delta is well-defined under this distribution but its inference is effectively dominated by `n_aligned_nonzero ≈ 6 + n_cross_nonzero ≈ 9 = 15`, not the surface n=60. Three odds-ratio-like quantities must be kept distinct. First, the pre-registered *median* odds ratio is formally $+\infty$ because the cross-slice median SMS is exactly 0; it degenerates under zero-inflation and licenses no verdict, so we report "aligned median > 0 = cross median" only as auxiliary qualitative evidence. Second, a post-hoc *detection-incidence* sensitivity, declared in its own single-test family (outside the pre-registered Holm family and outside the H2 pre-registration), binarizes SMS and asks how often each slice scores any nonzero effect. Under the MP5-held primary convention, aligned cells are nonzero in 6/12 cases versus cross cells in 9/48 (incidence 50% vs 18.75%; only 9 of the 12 PUTs carry any nonzero cell). A one-sided Fisher exact test gives $p = 0.036$ with conditional-MLE odds ratio 4.2 (sample OR 4.33; exact one-sided 95% CI $[1.12, +\infty)$), and the advantage is directionally stable across pool variants (OR 4.1–7.0, one-sided $p = 0.006$–0.049). Aligned meta-patterns therefore detect a nonzero effect appreciably more often than cross patterns, a distinct and modest incidence separation, not the near-twentyfold ratio that a PUT-level any-signal count (9 of 12 PUTs non-dead) would suggest if mislabelled as an aligned-cell count. Third, the Cliff's δ *magnitude* estimand (Section 5.3) is the pre-registered H2 criterion: incidence separates, magnitude does not, and the H2 verdict remains **not met**.

Effective-n note. The surface n_aligned = 12 and n_cross = 48 mask a much smaller information content: only about 15 observations carry nonzero SMS (6 nonzero aligned + 9 nonzero cross). This is a heuristic information count, not a formal effective sample size, but it explains the wide 95% bootstrap CI [0.014, 0.622] of the v4 primary delta (spanning negligible to large, consistent with the known liberal tendency of the percentile bootstrap when few observations carry signal). The implications for power and the H2 verdict direction are quantified jointly with the stipulated-alternative analysis in Section 5.4. PUT-class diversification at $n \geq 30$ is a testable route to relaxing the effective-n constraint. Power and effect-size disambiguation are treated jointly with the stipulated-alternative analysis in Section 5.4 (avoiding repetition).

H1 verdict: not met. H1 (RQ3) requires at least 4 of the 5 operators to produce ≥ 5 non-equivalent (confirmed) mutants on at least 9 of the 12 PUTs. Table 13 counts confirmed non-equivalent mutants per operator class across the 12 PUTs: only the Hyperparameter operator clears the $\geq 9/12$ bar.

Table 13. H1: PUTs reaching ≥ 5 non-equivalent mutants, per operator class.

Operator class	PUTs with ≥ 5 non-equiv mutants (of 12)
Conservation Erosion (CE)	4 / 12
Operator Substitution (OS)	5 / 12
Hyperparameter (HP)	**9 / 12**
Trajectory Flip (TF)	5 / 12
Structural Injection (SI)	1 / 12

Because the semantic operators are class-targeted, each is applicable to a subset of the four PUT classes rather than uniformly to all twelve. Only the broadly-applicable Hyperparameter operator clears the 9-PUT threshold. Thus 1 of 5 operators meets the criterion and **H1 is not met as pre-registered**; the uniform-applicability assumption behind the $\geq 9/12$ bar does not hold for class-targeted operators.

As a secondary interpretive denominator, we also count only PUTs for which the operator family has an explicit class-specific specialization in Appendix B.2 and the operator-level pilot records an attempted instantiation in Appendix C.4. Under this applicability-adjusted view, CE reaches 4/8 applicable PUTs, OS 5/7, HP 9/9, TF 5/6, and SI 1/6. The secondary denominator does not overturn the pre-registered H1 verdict, but it identifies the scientific source of the failure: HP and TF are implementable across their intended classes, OS is moderately broad, and SI remains a narrow high-risk family rather than a uniformly applicable operator. The per-operator implementability profile (r_{sem}, d_{impl}) is reported in Appendix C.4.

5.3 RQ4: Aligned and Cross-Pattern Comparisons

Table 14. RQ4: Aligned and cross-pattern comparisons.

Slice	n	Mean SMS	Median SMS
aligned ($j = k$)	12	0.213	0.100
cross ($j \neq k$)	48	0.077	0.000

The means and medians above are those of the MP5-held cross-source pool (`rq2_cliffs_delta_v4_mp5.json`) that generates the headline $\delta = 0.314$ reported below, so the summary statistics and the effect size share one provenance. (The unconditioned v4 pool with the c-class primary at MP1 gives higher means 0.275 / 0.061 and $\delta = 0.439$; the power analysis in Section 5.4 is run on that unconditioned pool, as noted there.)

Two delta point-estimates under the pre-registered primary-MP convention:

- **v3 (same-source, pre-registered):** delta = **0.323**, 95% CI [0.017, 0.622]
- **v4 (cross-source, c-class held at partial-order):** delta = **0.314**, 95% CI [0.014, 0.622]. The cross-source pool aggregates Claude / GPT / DeepSeek under an identical prompt; the c-class primary is held at the pre-registered partial-order meta-pattern so that the v3 → v4 contrast holds the prompt and primary-MP choice fixed. Because v4 uses mechanical gates rather than the v3 dual-blind reviewer step, the source-axis reading is qualified by the protocol-asymmetry analysis in Section 6.3.

H2 verdict: not met. H2 carries two pre-registered criteria, Cliff's $\delta \geq 0.474$ and an aligned-to-cross odds ratio ≥ 3.0. Neither delta crosses the Romano et al. [44] large-effect threshold 0.474, 0.323 and 0.314 both fall just below the Romano et al. [44] medium threshold of 0.330, so the delta criterion fails. The Romano threshold is retained as the frozen decision anchor because it is the threshold recorded in the experiment design; we do not re-anchor the verdict post hoc to a Vargha–Delaney interpretation scale. The odds ratio is degenerate: because the cross-slice median SMS is exactly 0 (table above), the median odds ratio is $+\infty$, so the ratio criterion is satisfied only trivially under zero-inflation and carries no evidential weight. H2 is therefore not met as a large-effect target. The contrast $\delta_{v4} - \delta_{v3} = -0.009$ (paired-role bootstrap 95% CI [-0.238, 0.207]) is too small to interpret as a stable LLM-source-diversity effect under the present asymmetric protocol: replacing a same-source pool with a three-LLM cross-source pool does not appreciably shift Cliff's delta within this design, but the exact source-diversity contribution requires a dual-blind v4 rerun. We frame H2 as an underpowered exploratory contribution and defer full verification to a follow-up study with a larger PUT sample.

Vacant-cell sensitivity. Excluding the 9 cells marked vacant in Table 6 gives the same H2 verdict. Under the frozen primary-MR convention, v4 has n = 11 aligned and n = 40 cross cells, $\delta = 0.323$ with 95% CI [-0.011, 0.648]; v3 has $\delta = 0.270$ with 95% CI [-0.043, 0.595]. The direction remains positive but still below 0.474, and the interval crosses zero once vacant cells are removed. The sensitivity is a denominator check, not a changed conclusion.

This medium-effect placement is consistent with Tip et al. [46] LLMorpheus's medium-effect range on JavaScript LLM mutants, a contextual literature observation (estimand caveat: their delta compares "LLM vs traditional mutants on fault detection", ours compares "aligned vs cross MP slice within one pool"; the numerical similarity is not substantive support).

5.4 RQ4: Stipulated-Alternative Power Analysis

The Section 5.2 effective-n constraint motivates an explicit power analysis. A plug-in bootstrap (5,000 replications, seed = 42) samples with replacement from the observed (n = 12, n = 48) v4 SMS distributions (the unconditioned pool with the c-class primary at MP1, mean SMS 0.275 / 0.061, $\delta = 0.439$; distinct from the MP5-held $\delta = 0.314$ pool that carries the primary H2 verdict). The bootstrap exceedance-probability table (probability that the resampled $\hat{\delta}$ exceeds each threshold under the observed distribution, not power against a stipulated alternative) is:

Table 15. RQ4: Bootstrap exceedance probability under the observed distribution.

Threshold	Interpretation	Exceedance prob. at (12, 48)
$\delta > 0.000$	Any effect	**0.997**
$\delta > 0.147$	Small	0.966
$\delta > 0.330$	Medium	0.759
$\delta >$ **0.474**	**Large (H2)**	**0.423**

The plug-in result answers the question "given the *observed* distribution, how often do we exceed the threshold?" A stipulated-alternative simulation answers a more pointed question: "if the *truth* equals the H2 boundary 0.474, how often does the (12, 48) design return $\delta \geq 0.474$?" SMS distributions have heavy ties at zero (45 of 60 cells), so a raw shift is discontinuous: any $\varepsilon > 0$ jumps δ from 0.314 to 0.74. We therefore use a mixture, drawing the aligned sample with probability w from (observed_aligned + 0.001) and with probability 1 − w from observed_aligned, calibrating w = 0.094 to realise $E[\delta] = 0.4746$.

Table 16. RQ4: Stipulated-alternative power.

Stipulated truth	Criterion	Power
0.474	$\hat{\delta} \geq 0.474$ (point estimate)	**0.491**
0.474	Lower 95% CI bound > 0 (any effect)	0.868

Even when the truth equals the H2 boundary, the point-estimate decision rule returns "not met" in roughly half of replications; a pass probability near one half at the boundary is intrinsic to a threshold rule applied to an approximately median-unbiased estimator at any sample size, so the 49.1% figure characterises the pre-registered decision rule, not a deficiency specific to the (12, 48) design, and its exact value depends on the stipulated alternative (the mixture with E[δ] = 0.4746 is one of many alternatives with that mean). This supports the framing in Section 5.3: the H2 verdict is a factual statement about the point estimate failing to clear the threshold, not a claim that the effect is necessarily smaller than 0.474. Increasing the sample size narrows the confidence interval but cannot lift the point estimate. The plug-in sample-size sweep (n_aligned ∈ {6, 12, ..., 60}; n_cross = 4 × n_aligned; power for $\delta > 0$ reaches 0.974 at n_aligned = 6 and 0.996 at 12, then plateaus; the small discrepancy with the 0.997 exceedance entry above reflects independent simulation runs) is in Appendix D.3.

The source-axis contrast is underpowered too. We did not run a separate power simulation for the source-axis contrast $\Delta\delta = -0.009$ (paired-role bootstrap 95% CI [-0.238, 0.207]); its estimand and dependence structure differ from the aligned-vs-cross δ, so the 49.1% figure above does not transfer to it. The small design (n = 12 PUTs) leaves the −0.009 null consistent with a wide range of true source-diversity effects, so we explicitly do not read it as evidence that source diversity is inert. A properly-powered strong-sense test is deferred to a follow-up study.

Note on monotone-transformation invariance. Cliff's δ is rank-based: a function of $U/(n_1 \cdot n_2)$, so applying a logit transform to SMS gives $\delta_{\text{logit}} \equiv \delta_{\text{raw}}$ by construction. This is consistent with the rank-invariance theorem and does **not** constitute additional robustness evidence. The genuine robustness threats to the H2 verdict come from the zero-mass dominance (Section 5.2), not from any metric-scale choice.

Table 17. RQ4: Root-cause diagnostic mass.

Metric	Value
Mean C1_share (v3 same-source pool)	0.164
Mean C1_share (v4 cross-source pool)	**0.209**
Mean suspect_share (v4 cross-source pool)	**0.791**
Cells with suspect_share $\leq$ 0.20 (secondary; threshold-calibration best, ood_band = 0.02)	12 / 60 = 20.0%

5.5 RQ4: Root-Cause Diagnostic Results

H4 verdict: not met. The pre-registered criterion is mean suspect_share ≤ 0.20; the observed mean is **0.791** (v4), decisively above the threshold, so H4 fails on its own metric. A dense cutoff sweep (Appendix D.2) confirms this is an intrinsic property of LLM-mutant pools rather than a calibration artefact: the secondary per-cell pass-ratio (cells with suspect_share ≤ 0.20) is flat at 12 / 60 = 20% across cutoffs $\in [0.05, 0.40]$, because v4's suspect_share distribution is severely bimodal (median 1.0, with 48 cross cells near 1 and 12 aligned cells near 0; the [0.20, 0.80] interior is nearly empty). H4 was pre-registered before pool characteristics were known, so this is a finding worth reporting.

C1-weighted sensitivity. Because LRCA deliberately does not filter SMS, we add a diagnostic secondary view $\mathrm{SMS}_{\mathrm{C1}} = \mathrm{SMS} \times \mathrm{C1\_share}$. On the v4 60-cell table, mean SMS is 0.104 and mean $\mathrm{SMS}_{\mathrm{C1}}$ is 0.0873; nonzero cells fall from 15/60 to 13/60. The conclusion is therefore conservative: suspect mass is high and H4 is not met, but the main positive SMS signal is not created solely by suspect labels. We keep raw SMS as the primary metric because LRCA is a diagnostic attribution layer, not part of the mutation-score denominator.

5.6 RQ4: Strong-Boundary Failure Modes

The 60-cell audit (Sections 5.2–5.5) exercises strong MRs in the regime where Theorem 3.5 holds. To map the strong boundary of Proposition 3.6 we run a complementary study on six third-party or textbook programs spanning the meta-patterns, kept separate from the 12 PUTs. Wherever a genuine fixed defect exists in a third-party library we use real version-switching: the pre-fix and post-fix releases run unmodified in isolated environments, and we record the repository, the version pair, and the issue/PR, and we fall back to a textbook faithful implementation only when the real defect cannot be exercised on the host (no `arm64` wheel, a heavyweight Fortran build, or a drift below machine resolution). $\varepsilon_{\mathrm{tol}}$ is an explicit factor in every case.

Table 18. RQ4: Strong-boundary map over six programs.

Family	Program (provenance; method)	Strong MR ($\varepsilon_{\mathrm{tol}}$)	Correct / faulty verdict	Boundary role
inv.con	non-conservative Burgers; [28] (textbook)	shock speed (0.03)	0.498 pass / −0.002 killed	strong
inv.eqv	scipy `align_vectors` 1.13.0 → 1.13.1, #20660 (real)	rotation residual (10^{-6})	10^{-16} pass / 2.0 killed	strong

adj.self	scipy `_adjoint` #8900/PR#8962 (real, extracted-diff)	adjoint-identity residual	10^{-16} pass / 1.4–2.0 killed	strong
rev.traj	leapfrog vs explicit Euler; [27] (textbook)	energy-drift ratio (≤ 3)	1.00 pass / 1952 killed	strong
PINN	soft vs hard BC; DeepXDE #26 (real torch)	exact BC $\lvert u(0)\rvert$ (10^{-4})	hard 0 pass / soft 5×10^{-4} killed	FP (weak MR)
struct. (FN)	numpy float32 RNG 1.21.6 $\rightarrow$ 1.22.0, #17478 (real)	mean / range / dist	both pass; buggy survives	FN (surviving)

Strong regime (four cases). For inv.con, inv.eqv, adj.self, and rev.traj the correct program satisfies the strong MR within ε_{tol} while the faulty program is killed, exactly as Theorem 3.5 predicts. Two are real fixed library defects (scipy `align_vectors`; scipy `_ScaledLinearOperator._adjoint`, whose pre-fix release has no `arm64` wheel and is therefore exercised by applying the exact fix diff in reverse); two are textbook contrasts standing in for a real defect that cannot be exercised on the host, a conservative versus non-conservative Burgers scheme for the GeoClaw `filpatch` conservation bug, and a symplectic leapfrog versus explicit Euler integrator for the REBOUND IAS15 energy drift, whose real signature near 10^{-14} lies below resolution.

False positive, weak MR (PINN). A physics-informed network that enforces the boundary condition as a soft penalty is a legitimately trained, non-faulty model, yet its exact-boundary residual $|u(0)| \approx 5 \times 10^{-4}$ exceeds a strict $\varepsilon_{\text{tol}} = 10^{-4}$, so the exact-boundary strong MR flags it: the discretisation (the soft constraint), not a fault, breaks the invariant. This is the discrete-program instance of the rotational-symmetry counterexample, a continuous invariant reproduced only approximately, and the empirical face of Proposition 3.6(i). The hard-constraint ansatz $u = x(1-x)\,\mathrm{NN}(x)$ satisfies the invariant by construction and passes, confirming that the violation is a property of the discretisation, not of training.

False negative, surviving non-equivalent mutant (RNG). The numpy float32 generator defect (#17478) zeroes the lowest mantissa bit of every sample; the structural MR family, mean ≈ 0.5, range $[0, 1)$, distribution shape, is satisfied by both releases within ε_{tol}, so the genuinely non-equivalent buggy version survives. This is coincidental satisfaction of the MR, and because the fault signature lies below the structural tolerance it is a tolerance fault; the surviving mutant is non-equivalent (an independent odd-bit-fraction discriminator reads 0.0 for the buggy release versus ≈ 0.5 for the correct one), and therefore distinct from a classical equivalent mutant. It is the empirical face of Proposition 3.6(ii).

The ε_{tol} trade-off. The PINN case makes the sensitivity–specificity coupling runnable: at the strict $\varepsilon_{\text{tol}} = 10^{-4}$ the legitimately-trained soft-BC model is killed (a false positive), while at a loose $\varepsilon_{\text{tol}} = 10^{-3}$ it survives correctly; if the same residual pattern instead arose from a true boundary-condition fault, loosening ε_{tol} would turn that survival into a false negative. Tightening ε_{tol} trades false negatives for false positives and loosening it does the reverse, as Proposition 3.6 requires. The strong boundary is therefore the honest scope of the duality, a tunable operating point, not a defect to be removed.

5.7 RQ4: Adjoint Extension Evidence

Where Section 5.6 maps the boundary at which the duality fails, a small confirmatory arm checks it holding in the forward direction for the sixth meta-pattern ψ_6. On the three third-party linear-operator libraries of Section 4.7 (scipy, pylops, jax), each carrying a real fixed adjoint defect, a strong adjoint MR detects only the active mutants: across all three substrates the correct program survives, every non-equivalent mutant (including the three real defects scipy #8900, pylops #292, and jax #6223) is killed, and no semantically equivalent mutant is killed, giving a forward SMS of 1.00 with zero false positives on the equivalent set (full subjects, mutant pools, and the per-substrate kill table in supplementary Appendix H). The arm is confirmatory and additive, it does not enter the 60-cell denominator and leaves H1–H4 unchanged, and carries two scope caveats: the scipy #8900 defect is shared with the Section 5.6 boundary study (so it is not counted as independent corroboration twice), and the pools are hand-constructed, low-cardinality sets with the real defects reproduced in extracted-diff form, so the arm demonstrates rather than measures adequacy (adjoint-arm reproduction-fidelity threat).

5.8 Cross-source contributes mutant quality, not effect size

Going from same-source to cross-source under the pre-registered primary-MP convention shifts Cliff's δ by only -0.009 (paired-role bootstrap 95% CI [-0.238, 0.207]). Cross-source pooling raises mean C1_share from 0.164 to 0.209 (a 27% relative increase), class-c mean SMS by **+91.5%**, and class-d mean SMS by 38.3%. Under an identical prompt template, three LLMs produce similar distributions for the aligned-vs-cross question. Within this fixed-prompt design, LLM source identity is therefore not the dominant factor behind the H2 ceiling: cross-source pooling improves the diagnostic quality of the kill set (LRCA C1 share, class-mean SMS) without moving the aligned-vs-cross effect size. This is a scoped null, not a general claim that LLM diversity is inert. The strong-sense source-diversity test, with per-LLM differential prompts (V_persona, V_cot), is deferred to a follow-up study. Appendix C.1 records the full protocol.

The Section 4.5 evidence (5.14% AST overlap, HP/SI/TF at 0/0/0) confirms that the medium-effect ceiling is not an artefact of LLM-pool overlap with the syntactic-mutant space. 94.86% of v4 mutants are AST-disjoint from cosmic-ray defaults, and the three classes unreachable under default first-order configurations (HP, SI, TF; 159 of 292 mutants; HOM not refuted, see Section 4.6) lie outside that space by construction. A larger aligned-versus-cross effect therefore points first to MR-design refinement. A larger sample would mainly reduce uncertainty around this design unless the MR design itself changes.

A numerical-coincidence note. Petrovi'c et al.'s [2021] ~20% productive-mutant rate at Google is close to our calibration-sweep C1_share of 0.20 (the threshold-calibration best of Appendix D.2, distinct from the v4 cross-source pool value 0.209 in Table 17 and the v3 same-source pool value 0.164). Their construct is a developer survey (subjective usefulness), and LRCA's C1 is a diagnostic annotation over observed kill behaviour; the agreement between the two is contextual, not mechanism validation.

5.9 Decoupling between Semantic Feasibility and Kill Rate

The operator-level pilot in Appendix C.4.1 reveals a sharp pattern: HP, TF, and OS operators on the c-class (surrogate) and d-class (machine learning) PUTs reach R_sem $\approx$ 1.0 (semantic feasibility, the LLM successfully produces a syntactically valid, executable, intent-bearing mutant) but R_kill = 0 under the PUT's primary MP (the AVP fails to detect the mutant under that MR). Concretely, six cells show R_sem $\geq$ 0.9 with R_kill = 0: c1_CE1 (GPR `noise_level` 1e-4 $\rightarrow$ 1e-1), c2_OS1 (polynomial $\rightarrow$ spline basis), c3_HP1 (relu $\rightarrow$ tanh), c3_TF1 (max_iter 1000 $\rightarrow$ 5), d1_HP1 (MLP α

1e-4 → 1.0), and d3_HP1 (LR C 1.0 → 1e-4). In each, the mutant *is* a valid hyperparameter-semantic injection, but the primary MP, partial-order (asymptotic) for the c-class and monotonicity for the d-class, is an asymptotic or statistical relation, and the parameter deviation it tolerates is wide enough to absorb the mutant. We note a family-boundary caveat these examples expose: edits such as an activation-function swap or an iteration-cap cut sit near the OS/HP and HP/TF boundaries, and the family label records the generating operator's intent rather than a unique structural class; the aligned-versus-cross reading should be taken with this labelling slack in mind.

S5 stratum-purity audit. We verified S5 directly rather than resting on generation intent. Running all five invariant checkers on each of the 292 admitted mutants (the deterministic offline AVP dispatcher, 20-repeat majority vote; SSOT `s5_purity_v4.json`) yields the invariant-flip distribution $\{0\!:\!170,\ 1\!:\!93,\ \geq 2\!:\!29\}$. The effect map σ is therefore single-valued on 263/292 (90.1%) of the corpus; the 29 multi-stratum mutants (9.9%) are confined to two operator families that mutate shared upstream state, control-flow reversal (CF, 9/9) and training-data corruption (TF, 20/54), while the four local-edit families (CE, OS, HP, SI) are 100% pure (0/229; Table 19). Re-attributing the aligned-versus-cross off-diagonal kill mass, 57/88 (64.8%) comes from pure single-invariant mutants (genuine cross-stratum detection) and 31/88 (35.2%) from multi-stratum mutants, the latter confined to the four cells `B2_MP1`, `C1_MP1`, `C1_MP2`, `D1_MP5`, and `D3_MP5`. The fiber partition thus survives at the σ level (R2's $\geq$ 90%-purity decision rule is met), and the aligned-versus-cross contrast should be read as a 57:31 pure/multi-stratum split rather than as uniformly pure cross-stratum detection.

Table 19. S5 stratum-purity audit of the v4 corpus. "Detected" = killed by at least one MP invariant checker; "pure" = killed by exactly one (S5 holds); "multi-stratum" = killed by two or more (σ multi-valued). Multi-stratum leakage is confined to CF and TF, which mutate shared upstream state.

Operator	Mutants	Detected	Pure (flip 1)	Multi-stratum ($\geq$ 2)
CE	64	27	27	0
OS	60	35	35	0
HP	72	11	11	0
SI	33	11	11	0
CF	9	9	0	9
TF	54	29	9	20
All	**292**	**122**	**93**	**29**

The cell-level evidence in Section 5.2 reproduces this pattern: 75% of cells are zero, concentrated in the cross slices. Under v4 cross-source, mean C1_share rises from 0.164 to 0.209 (+27%): among the few killed mutants, the cross-source pool produces cleaner LRCA diagnostic labels. Source diversity therefore improves the *quality* of the kill set without expanding its *coverage*. **The engineering insight is that operator-MP alignment in MR design is necessary for strong SMS signals; merely enlarging the semantically feasible mutant pool dilutes the C1 proportion without lifting the kill rate.** Two open questions remain: whether SMS can infer which MP class is missing for which PUT class, and whether LRCA can be calibrated with class-specific thresholds separating likely semantic failures from artefacts.

This decoupling motivates the source-axis reading in Section 5.3: the v3 → v4 micro-shift of −0.009 estimates LLM source diversity under a fixed prompt, while MR-design adequacy at the operator-MP level governs the kill rate in this design.

5.10 RQ4: Industrial Construct-Separation Evidence

The industrial arm answers the last clause of RQ4: whether the construct separation among aggregate kill-rate, semantic alignment, and real-defect detection remains visible outside synthetic mutants and outside author-written kernels. The arm is a 34-case study over reproduced, MR-detectable defects from widely used scientific-computing libraries, drawn from an archived dataset [38] (10.5281/zenodo.21203424); each case pairs a buggy and a fixed revision and registers the metamorphic relation that discriminates the two. We report verified result totals only, and the arm does not change the paper into a benchmark article: defect mining, rejected cases, repository tooling, and benchmark release are left outside the paper and remain contributions of the dataset. The result-level evidence summary used here is included in the supplementary evidence package: it records the five-stratum counts, the aggregate mutation-phase statistics, the 34/34 real-defect face, and the non-nesting case identifiers, but not candidate mining or tooling. In the four-group mutation-phase comparison pre-registered in the dataset protocol, the four groups are the case's registered pattern-derived relation (T1), literature-generic baseline relations (B1), seeded random relations (B2), and mechanical ablations of T1 (A1); the comparison family is three one-sided case-level Wilcoxon signed-rank comparisons (T1 > B1, T1 > A1, B1 > B2) under Holm correction, and the case census (n = 34) was fixed by the dataset's verification status before any comparison was run. Under the primary all-mutant denominator the group kill rates are T1 377/1124 = 0.335 (Wilson 95% CI [0.308, 0.364]), A1 348/1124 = 0.310, B1 274/1124 = 0.244, and B2 228/1124 = 0.203. The pattern-derived group beats the literature-generic baselines (mean paired difference +0.101, bootstrap 95% CI [+0.029, +0.179], Holm-adjusted $p = 0.046$, Cliff's $\delta = +0.247$; one-sided Wilcoxon signed-rank $V = 279.5$, $z = 2.16$, unadjusted $p = 0.015$; case-resampling bootstrap $B = 10{,}000$, seed 20260704). Because the Holm-adjusted p sits close to the 0.05 boundary, we additionally report distribution-free checks of the *same* one-sided $T1 > B1$ contrast (robustness of the pre-registered estimand, not new hypotheses): an exact sign-flip permutation test on the signed-rank statistic (full 2^{27} enumeration over the 27 nonzero paired differences) gives $p = 0.014$, a Monte Carlo sign-flip test on the mean paired difference ($B = 10{,}000$) gives $p = 0.005$, and the BCa bootstrap 95% CI for Cliff's δ ($B = 10{,}000$) is $[+0.07, +0.46]$, excluding zero. The verdict is marginal under the Holm-adjusted normal approximation but robust under exact inference, and is reported with its fragility: the paired difference narrowed when the dataset grew from 30 to 34 cases, and a sensitivity rerun recorded in the dataset report (excluding the one case with a known tolerance-storm pathology) leaves the Holm-adjusted verdict unchanged. The other two family members are not significant after Holm correction: the pattern-derived group does not significantly dominate its ablations, and the generic baselines do not significantly dominate the random draws. On the real-defect face, however, the distinction becomes sharper: the pattern-derived relation detects every real defect in the reported mutation-suite face (34/34). Because case admission itself requires an MR-detectable behavioural difference under the registered pattern-derived relation, this 34/34 count is selection-conditioned rather than evidential; the informative content of the face lies in the contrast on the same admitted cases: the literature-generic baselines detect nothing in 27 of 34 cases, the seeded random relations in 26 of 34, and the dimension-reduction ablation loses the real defect in 19 of 34 cases that its parent relation detects; a second, de-strictification ablation loses it in 17 of 34, with 11 cases lost by both (supplementary Appendix I).

The arm's cases were selected by a fixed inclusion rule: a case had to have a verified semantic fault, an executable pre/post or faithful contrast, and an MR-detectable behavioral difference under the declared stratum. Cases requiring unsupported build stacks, lacking an executable contrast, or failing MR-detectability verification were excluded before the reported mutation-phase comparison. No case was dropped after observing the comparative SMS or baseline detection results. The face is therefore a selection-conditioned, face-level check on construct separation, not a corpus-level claim about real-defect coverage.

This pattern is consistent with the construct separation predicted by the fiber view of Section 3.3.9. A mutant kill-rate is evidence about which sampled semantic effects an MR observes; it is not, by itself, a verdict on MR quality. The same industrial arm gives concrete non-nesting counterexamples, A-LAPACK-004, A-OPENBLAS-001, B-POCKETFFT-002, and E-ORDINARYDIFFEQ-001, where generic baselines kill mutants that the pattern-derived relation can miss or match. The mechanisms are distinct across the four cases, value destruction that preserves element order, zeroing that preserves a symmetry the relation checks, overshoot that stays within the tested bounds, and a variable-substitution/stale-read edit that operator mutation does not reach well, so no single relation family induces a total order over the others (supplementary Appendix I). Thus semantic mutation should be read as a certificate-bearing diagnostic for a declared semantic stratum, not as a scalar hierarchy in which one mutation family simply dominates another. This evidence therefore strengthens the paper's main argument while keeping benchmark construction outside the manuscript's contribution boundary.

5.11 RQ5: Cross-Class Consistency

Table 20. RQ5: Cross-class consistency and Friedman statistics.

Class	Mean SMS (v3)	Mean SMS (v4)
a (numeric)	0.067	0.067
b (probabilistic)	0.156	0.148
c (surrogate)	0.047	**0.089 (+91.5%)**
d (ML)	0.081	0.112 (+38.3%)

Sign test (within-class aligned mean − cross mean, sign = +): **3 / 4 (partial) under both v3 (same-source) and v4 (cross-source)**; the b-class is the inverted cell.

H3 verdict: not met. H3 carries two pre-registered criteria, a within-class sign test of 4/4 and CV(ΔSMS) < 0.5. The sign test is 3/4 (partial) under both v3 and v4, cross-source pooling does not flip the b-class inversion, so the consistency criterion fails. The per-class effects ΔSMS_c (aligned mean − cross mean) span from negative in the inverted class to large positive elsewhere, giving CV(ΔSMS) well above 1, far exceeding the 0.5 dispersion threshold, so the dispersion criterion also fails.

The mixed-effects primary model `sms ~ C(class) * C(operator) + (1 | put)` returned Singular matrix; the fallback `sms ~ C(class) + C(operator) + (1 | put)` had PUT random-intercept variance hit boundary 0 (degenerate). We therefore use Friedman as the non-parametric formal alternative:

- **PUT (n=12) × MP (k=5): chi^2 = 16.76, p = 0.0022** (significant; exploratory inferential per Table 11; v4 pool, `rq3_friedman_v4.json`).
- MP rank means: 3.08, 2.58, 2.00, 3.00, 4.33.

- Per-class Friedman with **Bonferroni × 4 correction**: a 1.000 / b **0.140** / c 0.924 / d 1.000, no per-class result remains significant after correction. Kendall's W effect sizes: a 0.333 (small) / b 0.862 (large concordance, but caveat: N=3 per class makes this label nominal only) / c 0.467 / d 0.417.

Caveat. Friedman tests "are MP rank differences present" (averaged over PUTs); H3 tests "is direction consistent across 4 classes". These are logically independent. H3 stands on the Section 5.11 sign test; per-class Friedman is descriptive only (small N=3 per class).

5.12 RQ5: Semantic-Mutation Score and Pattern Coverage

Status. The SMS–Pattern-Coverage relationship, a descriptive sub-question of RQ5, is reported as a descriptive observation, not a hypothesis test, because n = 12 places the 95% Spearman CI at roughly $[-0.5, +0.6]$, so the test cannot distinguish zero, moderate-positive, or moderate-negative correlation. The numbers below are recorded so that a future study (n ≥ 30 PUTs) can pre-register a directional hypothesis.

Pattern Coverage (PC) per PUT = #triggered (MP_k, R_outcome) cells / 10. Range [0.500, 1.000], mean 0.733. Pairing with mean SMS over 5 MPs: Spearman rho = **0.163**; Kendall tau = 0.136; n = 12 (descriptive; no inferential reading attaches).

Statistical-power caveat first. At n = 12, Spearman's 95% CI is approximately [-0.5, +0.6] (Fisher-z approximation, a rough small-sample heuristic); the test cannot distinguish "zero correlation" from "moderate positive correlation" or "moderate negative correlation". Consistent with the descriptive licence of this row (Table 11, "none"), we attach no p-value; the descriptive ρ and τ do **not** support the strong claim that "SMS is independent of PC" or the strong claim that "SMS is strongly correlated with PC". The conservative finding is *no detectable correlation at n = 12*; orthogonality is a hypothesis for a future study (n ≥ 30 PUTs or refined PC operationalisation; full PC operationalisation in Appendix D.5).

5.13 Class-Level Stability and Source-Axis Interpretation

Section 5.11 gives the formal H3 verdict, so this deployment-facing reading is deliberately shorter: cross-source pooling improves the c- and d-class means (+91.5% and +38.3%), consistent with c / d classes having higher mutant-diversity demand than a / b, but it does not repair the b-class inversion or the CV(ΔSMS) dispersion failure. The mixed-effects singularity and the Friedman fallback are treated once, in that RQ5 result section, not as a second independent result here.

5.14 Practical Interpretation and Deployment Scope

Scope. All deployment claims in this subsection are bounded to single-output `float` → `float` kernels under 2 KB, the 12 PUTs of this paper. They do not apply to industrial-scale, multi-module, or multi-output scientific computing software, which lies beyond the scope of this paper. Table 21 gives interpretive readings for SMS and LRCA outputs; it is not an acceptance-threshold table.

Table 21. Reading SMS and LRCA together.

Observation	Interpretation	Design response
Low SMS, high C1_share	The MR family detects few mutants, but the kills it does make are mostly semantic	Add MRs for adjacent semantic strata before enlarging the mutant pool
Low SMS, high suspect_share	The MR family mainly triggers tolerance, OOD, statistical, or artefact labels	Inspect tolerances, input domain, and statistical assumptions before treating kills as adequacy evidence
High SMS, low real-defect detection	Mutant kill-rate and defect detection have diverged	Revisit the declared semantic stratum and add real-defect-facing MRs
Generic relation kills outside the pattern-derived relation	Relation families are non-nested	Treat the result as a different semantic-effect fiber, not as domination by one MR family

Test engineers can read SMS as a per-MR scalar adequacy score. When SMS sits well below the aligned baseline of about 0.21 (Section 5.3), the LRCA labels, C2 tolerance, C3 OOD, C4 statistical, point to specific repair paths. Air-gap incompatibility is a hard limitation: the workflow depends on external LLM API calls and is incompatible with most regulated air-gapped Verification and Validation (V&V) environments (IEC 60880, DO-178C, IEC 62304, ISO 26262). Pre-generated mutant pools can be reproduced offline, but generating new mutant pools requires LLM access. Self-hosted open-weight LLMs and offline-cached pools are future mitigations. Appendix E.1 gives the full air-gap justification and the standards catalogue.

MR designers can use offline batch SMS runs as a quantifiable design-feedback metric. We recommend a quarterly batch audit (about 0.5 person-day per quarter; detailed cost breakdown in Appendix E.2; estimates based on observed timings during this paper's 12-PUT campaign) rather than per-pull-request gating, because LLM API latency, cost non-determinism, and air-gap incompatibility together rule out the per-pull-request style. Appendix E.2 gives the quarterly-audit workflow with resource-cost table.

V&V documentation can carry SMS as research-grade supplementary evidence alongside code coverage and MR lists. We make no normative claim toward IEC, ISO, or ASME standards. An earlier draft proposed acceptance thresholds (aligned-cell SMS ≥ 0.20 or 0.30 plus C1_share ≤ 0.20); we removed them in revision because they have no normative backing and could be misread as enforcement-ready. Appendix E.3 records the conceptual complementarity with the code-verification scope of ASME V&V 20-2009 §3.

All three stakeholder classes consume the same single source of truth (`paper_numbers_v4.json` and `lrca_60cell_v4.json`) to avoid documentation fragmentation.

Example MR-design use. If a surrogate-model PUT has R_sem ≈ 1 for a Hyperparameter mutant but SMS = 0 under the primary monotonicity MR, the design action is not to generate more mutants of the same kind. The SMS / LRCA reading is that the MR family is blind to the admitted semantic-effect fiber; the next design step is to add a convergence, stability, or fidelity-order MR for that PUT before claiming adequacy for the hyperparameter stratum.

6 Threats to Validity

Table 22. Threats to Validity.

Threat area	Threat	Class	Mitigation summary	Detail
LLM generation	LLM generation reproducibility	Internal	Archived raw responses + frozen mutant pools (no user-facing seed; reproducibility by archival)	Appendix F.1
Equivalence approximation	E2 probabilistic approximation	Construct	K_eq = 1000 + Hoeffding bound; sweep deferred to future work	Appendix F.1
Shared infrastructure	Shared-infrastructure dependency	Internal	AVP version pinned to the upstream commit hash; embedded source	Appendix F.1
Root-cause diagnosis	LRCA multi-label boundary	Internal	Decision-tree priority + multi-label co-occurrence table	Appendix F.1
Representativeness	12-PUT representativeness	External	shared infrastructure; Appendix B.1 coverage argument; scaling to industrial PUTs	Appendix F.2
Statistical power	Cross-class statistical power	Conclusion	H3 framed exploratory; Friedman fallback; mixed-effects unavailable	Appendix F.2
LLM homogeneity	LLM homogeneity bias	Construct	3-LLM rotation + 20% manual sampling	Appendix F.2
LLM-source shift	LLM-source distributional shift	External	DeepSeek 11/15 of overlaps; argument is categorical, not frequency	Appendix F.2

Pool size	Mutant-pool size	Internal	v3 mean 17.4, v4 mean 24.3 per PUT; effect intrinsic, not pool-dilution	Appendix F.1
LLM non-determinism	LLM non-determinism	Internal	De-dup, K = 10/20 repeats, raw-response store	Appendix F.1
Higher-order mutation	HOM equivalence	External	Confined claim to first-order syntactic tools; HOM testing deferred	Appendix F.2
Protocol asymmetry	v3 vs v4 protocol asymmetry (dual-blind reviewer)	Internal	Quality not down (C1_share 0.164 → 0.209); follow-up to rerun dual-blind on v4	Appendix F.1
Adjoint reproduction	Adjoint-arm reproduction fidelity and by-construction pools	Internal	Real defects reproduced through extracted diffs (no `arm64` wheel); arm is confirmatory on hand-constructed, low-cardinality pools, not a high-adequacy discovery; scipy #8900 is shared with the Section 5.6 boundary study	Section 5.7
Real-defect evidence	Industrial real-defect arm boundary	External	Used at result level to confirm construct separation; reusable benchmark construction deferred outside this manuscript	Section 5.10

Final limitations. We list ten known limitations.

(1) **Equivalence determination is a probabilistic approximation.** Sampling K_eq = 1000 inputs is an engineering implementation of the undecidable equivalence problem, not a theorem-based decision. A Hoeffding-style upper bound on the false-equivalence probability is given in supplementary Appendix F.1, but we did not execute a K_eq sweep over {500, 1000, 2000} for this submission. A follow-up study will run that sweep.
(2) **LLM generation homogeneity bias.** Even with three-LLM cross-source pooling, training-data overlap across Claude, GPT, and DeepSeek may produce similar blind spots. Three-LLM rotation and 20% manual sampling reduce but do not eliminate this risk. A stronger replication should add a fourth, independently trained LLM family; a self-hosted open-weight model is the natural candidate.
(3) **Limited statistical power for cross-class consistency.** The four-class sign test has df = 3, and the mixed-effects model returns Singular matrix at N = 60 over 12 PUTs. We treat H3 as exploratory and present RQ5 in four pieces: class means, sign test, Friedman, and a forest plot.
(4) **LRCA reports "likely" root causes.** The decision-tree priority C5 > C4 > C3 > C2 > C1 is an engineering choice. The reproducibility package preserves the multi-label co-occurrence table for every killed mutant as well as the priority-winning root cause; `root_cause` is best read as a likely cause rather than a definitive causal attribution.
(5) **The AVP is reused from the shared infrastructure.** Our reproducibility package embeds the AVP source code so that the package stays self-consistent under upstream evolution, but interface semantics may shift if the upstream infrastructure undergoes a major revision.
(6) **Epistemological scope versus engineering scope.** SMS measures semantic detection capability in the epistemological sense; engineering value is not evaluated here.
(7) **Signature simplification.** The `float → float` single-output signature is a substantive constraint, not a purely engineering trade-off, and it bounds the upper limit of mutant semantic complexity. A portability study should validate SMS on industrial-grade multi-output PUTs.
(8) **Air-gap incompatibility.** The mutant-generation workflow calls external LLM APIs and is incompatible with most regulated air-gapped V&V environments (IEC 60880, DO-178C, IEC 62304, ISO 26262). Pre-generated mutant pools can be reproduced offline, but generating new pools requires LLM access. Self-hosted open-weight LLMs and offline-cached pools are future mitigations.
(9) **No proof-carrying semantic-effect certificates.** The present workflow admits mutants through E1 ∧ E2, LRCA classification, and AST-normalised audit trails. These are empirical and structural certificates, not machine-checkable proof objects. In particular, we do not yet define an explicit abstraction function from concrete program semantics to scientific invariant domains, do not parameterise invariant families such as conservation, convergence, and stability as abstract-domain elements, and do not attach per-mutant proof, error expression, or proof-label certificates that bind a syntactic edit to a declared semantic-effect class. This is the main gap between SMS and a certificate-bearing semantic mutation framework.
(10) **Real-defect evidence is used narrowly.** The industrial real-defect arm strengthens this paper's construct argument by showing that mutant kill-rate, semantic alignment, and real-defect detection can separate in practice on reproduced library defects. We do not claim benchmark construction, candidate curation, repository tooling, or corpus release as contributions of this paper; those remain contributions of the archived dataset the arm draws on. The semantic-mutant SMS instantiation itself is evaluated on the 12 controlled kernels only; extending SMS computation to the industrial cases is registered follow-up work on the same dataset.

6.1 Statistical-Modelling Capacity

The mixed-effects primary model `sms ~ C(class) * C(operator) + (1 | put)` returned a Singular matrix at N = 60; the fallback `sms ~ C(class) + C(operator) + (1 | put)` had the PUT random-intercept variance hit boundary 0 (degenerate). Both failures are not numerical accidents; rather, they are evidence that 60 observations across 12 PUTs cannot identify an 11-dimensional fixed-effects structure plus 12-PUT random intercepts. This is a hard limit of the present design: **the experiment cannot, in principle, fit a hierarchical model with this fixed-effect dimensionality at this sample size**. Friedman tests serve as the non-parametric formal alternative, but they answer a structurally different question. Expansion to n_PUT ≥ 30 is the only path to a properly identified hierarchical model.

6.2 Programs with Zero Observed Signal

Figure 2 and Section 5.9 document that PUTs A1, A3, and C2 produce all-zero rows across the five MPs in the cross-source pool; that is, **for 25% of the PUTs, SMS gives zero signal at every operator-MP cell**. This is a substantive limitation of SMS as an adequacy metric on this PUT cohort: when the LRCA C1 mass collapses to zero across all aligned and cross slices for a PUT, SMS cannot distinguish strong from weak MR sets on that PUT. The phenomenon coexists with R_sem ≈ 1.0 (mutants are valid; Section 5.9), so the issue is not mutant generation but MR-design adequacy at the PUT level. MR-design refinement (per-PUT MP discovery) and the proposed pre-registered c-class primary-MP rule on a fresh dataset are the principal mitigation paths. We declare this as a residual PUT-level zero-signal threat.

6.3 Protocol-Asymmetry Magnitude Estimate

Section 4.9 declared protocol asymmetry: v3 used the Phase-1 dual-blind reviewer protocol (Claude generation + GPT-5.4 review + DeepSeek arbitration), while v4 passes V1-V4 mechanical gates only. To quantify the potential δ-shift attributable to this asymmetry without dual-blind, we offer a **rough order-of-magnitude estimate**: in the Phase-1 audit, dual-blind review filtered approximately 5-10% of LLM-generated mutants as semantically inconsistent; if these filtered mutants were systematically less effective at killing under MR_aligned (a plausible but unverified premise), the upper-bound δ-shift contribution from removing the reviewer step is bounded above by approximately **±0.03 to ±0.05 on Cliff's δ** at n_aligned = 12. This bound is an order of magnitude larger than the observed −0.009 source-axis shift and therefore prevents reading the source-axis contrast as evidence of LLM-source-diversity inertness; a direct dual-blind v4 rerun (deferred to a follow-up study) would give a tighter quantification. Such a rerun on the same 12 PUTs only removes the protocol-asymmetry confound: at $n = 12$ the $\Delta\delta$ CI stays ≈ 0.44 wide, so detecting a $\Delta\delta = 0.20$ effect at 80% power requires ≈ 30 PUTs, and the follow-up is planned at $n \geq 30$.

7 Future Work

Seven directions follow from this study.

1. A pre-registered c-class primary-MP rule on a fresh dataset.
2. A differential-prompt LLM-diversity test using V_canonical, V_persona, and V_cot (full protocol in Appendix C.1.1).
3. A scaling study at n ≥ 30 PUTs for SMS-vs-Pattern-Coverage orthogonality and for HOM-equivalence empirics.
4. An abstract-semantics certificate layer: define the abstraction α from concrete executions to scientific invariant domains, specify CE/OS/HP/TF/SI as semantic-effect predicates over those domains, and attach per-mutant proof objects, error expressions, or proof labels that certify class membership.

(5) A rerun of the dual-blind reviewer protocol on the full v4 grid, to separate protocol asymmetry from source diversity.
(6) Cross-language portability work (Python → JavaScript and Julia) building on Tip et al. [46].
(7) A self-hosted open-weight LLM pilot for air-gapped industrial deployment.

8 Conclusion

8.1 Findings summary

The empirical study supports the semantic-mutation construct in seven ways. The pre-registered thresholds are treated as stress tests of this instantiation, not as prerequisites for the SMS definition.

(1) SMS preserves the classical MS ratio while moving the operative definitions of mutant generation, equivalence, and killing into declared semantic strata. The 60-cell audit therefore evaluates how one concrete MR design observes semantic-effect fibers, rather than redefining mutation testing from scratch.
(2) The structural audit separates the semantic and syntactic mutant spaces. Across 12 PUTs, semantic mutants have 5.14% AST-normalised overlap with default cosmic-ray syntactic mutants; Hyperparameter, Structural Injection, and Trajectory Flip are 0/72, 0/33, and 0/54 under first-order defaults.
(3) Operator-MP alignment produces a positive but not large effect in this design. The H2 large-effect threshold is **not met under the pre-registered point-estimate criterion**: $\delta_{v3} = 0.323$, just below the Romano et al. [44] medium threshold of 0.330. The stipulated-alternative analysis shows the point-estimate rule itself passes with probability near one half when the truth sits at the boundary, so the verdict is a point-estimate result, not a proof that the true effect is below the large-effect threshold.
(4) Cross-source pooling changes mutant quality more than aligned-versus- cross separation. Holding the c-class primary metamorphic relation at the pre-registered partial-order meta-pattern, three-LLM pooling under an identical prompt shifts Cliff's δ by only -0.009 (paired-role bootstrap 95% CI [-0.238, 0.207]), while raising mean C1_share by 27%, class-c mean SMS by 91.5%, and class-d mean SMS by 38.3%.
(5) The MP structure is visible, but cross-class consistency is not yet stable. The Friedman main effect on MP differentiation is $\chi^2 = 16.76$, $p = 0.0022$, whereas the pre-registered H3 criteria are not met: the class-level sign test is 3/4 and CV(ΔSMS) exceeds 0.5. The Spearman correlation between SMS and Pattern Coverage is 0.163 at $n = 12$, so orthogonality remains a future hypothesis.
(6) The industrial real-defect arm supports construct separation beyond the small PUT audit. Pattern-derived relations detect all 34 tabled real defects in the reported face, while the pre-registered four-group comparison shows only a modest aggregate kill-rate advantage over literature-generic baselines, and non-nesting cases remain in both directions. Thus kill-rate, semantic-stratum alignment, and real-defect detection should be interpreted as related but distinct constructs, on industrial code and not only on the controlled kernels.
(7) The remaining failed thresholds delimit the current instantiation. Operator implementability H1 is not met because only Hyperparameter reaches ≥ 5 non-equivalent mutants on $\geq 9/12$ PUTs, and LRCA-mass H4 is not met because mean suspect_share is 0.791. These results identify where MR design and attribution need refinement; they do not change the degeneration theorem or the strong-boundary analysis.

8.2 Methodological contributions

This paper contributes a three-layer framework for domain-semantic mutation in single-output scientific computing kernels.

- **Layer 1 (Section 4.2).** Formal necessary conditions for semantic mutation, cross-function-boundary substitution, dependence on domain knowledge, change in algorithmic class, instantiated as five operator families: CE, OS, HP, TF, and SI (also written mut_C, mut_M, mut_G, mut_T, mut_F).
- **Layer 2 (Section 3.3.3).** The E1 $\wedge$ E2 equivalence judgement, the conservative complete instantiation of the Layer-1 conditions, with an explicit trade-off against E1-alone and E2-alone variants.
- **Layer 3 (Section 4.5).** AST-normalised empirical traceability across all 12 PUTs: a 5.14% overall AST overlap with cosmic-ray defaults, and HP, SI, and TF unreachable at 0/0/0 under default first-order configurations (HOM not refuted; see Section 4.6).

SMS is backward-compatible with the classical MS through Theorem 3.1 (Section 3.3.6), which establishes almost-everywhere degeneration in the limit $L = L_{\text{equiv}} \wedge L_{\text{killed}} \wedge L_{\text{mut}}$. The 60-cell empirical audit reported in Section 5 is one demonstration following this backbone, not the paper's main contribution.

Data and Code Availability

All raw data, JSON SSOTs (paper_numbers_v3 / v4, rq2_cliffs_delta_v3 / v4_mp5, rq3_friedman_v4, s5_purity_v4, h2_incidence_v4, lrca_60cell, lrca_v4_mp5_recompute, rq2_power_stipulated, cosmic_ray_12put_ast_diff), mutant pools, AVP source, and analysis scripts are archived on Zenodo under DOI 10.5281/zenodo.20250664. The industrial real-defect arm (Section 5.10) draws its result-level statistics from a separately archived defect dataset maintained by the authors [38], available on Zenodo under DOI 10.5281/zenodo.21203424; benchmark construction and governance are contributions of that dataset, not of this paper. Every industrial (RQ4) number is nonetheless recomputable in-repo: the per-case 34-case × {T1, B1, B2, A1} kill matrix and per-case real-defect face are mirrored as a repository SSOT at `data/results/industrial_percase_v1.json` (provenance DOI 10.5281/zenodo.21203424 with archive SHA-256 recorded in the file), from which `scripts/build_industrial_ssot.py` re-derives all group kill rates, the paired-comparison family, the real-defect face counts, and the robustness battery, failing the build on any mismatch with the manuscript. An earlier version of this work is available on arXiv: arXiv:2605.17437 [cs.SE]. For peer review, a read-only mirror of the repository can be provided by the corresponding author upon Editor request, per journal guidelines. The repository structure follows `REPRODUCIBILITY.md` in the source tree. The cosmic-ray and mutmut operator versions referenced in Section 4.5, Section 4.6, and Appendix B.6 are pinned in `requirements-frozen.txt`.

Supplementary Material

Appendices A–I (notation and operator catalogue; experimental subjects and operator specialisations; experimental procedure details; statistical analysis details; deployment considerations; detailed threats-to-validity mitigation; the full SMS-to-MS degeneration proof; the adjoint extension arm; and a result-level real-defect evidence summary) are provided as separate online supplementary material.

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CRediT authorship contribution statement

Meng Li: Conceptualization, Methodology, Software, Writing, original draft. **Xiaohua Yang**: Supervision, Formal analysis, Writing, review and editing. **Jie Liu**: Investigation, Validation. **Shiyu Yan**: Data curation, Visualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Generative AI Disclosure

The authors used AI tools under human direction for editorial proofreading, LaTeX maintenance, consistency checks, and review-simulation support. All research claims, methods, experimental data, analyses, and conclusions were authored, verified, and approved by the human authors.

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